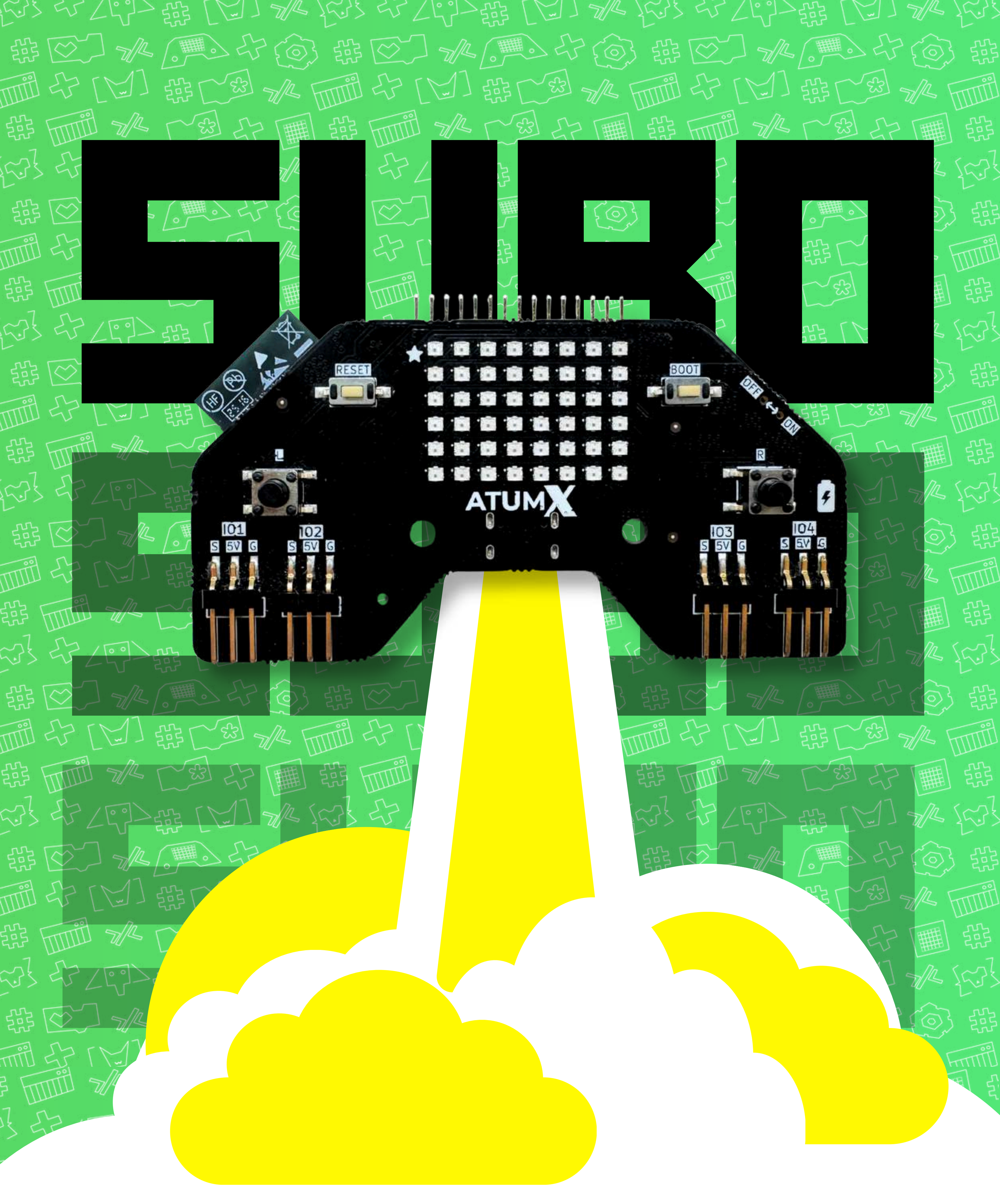
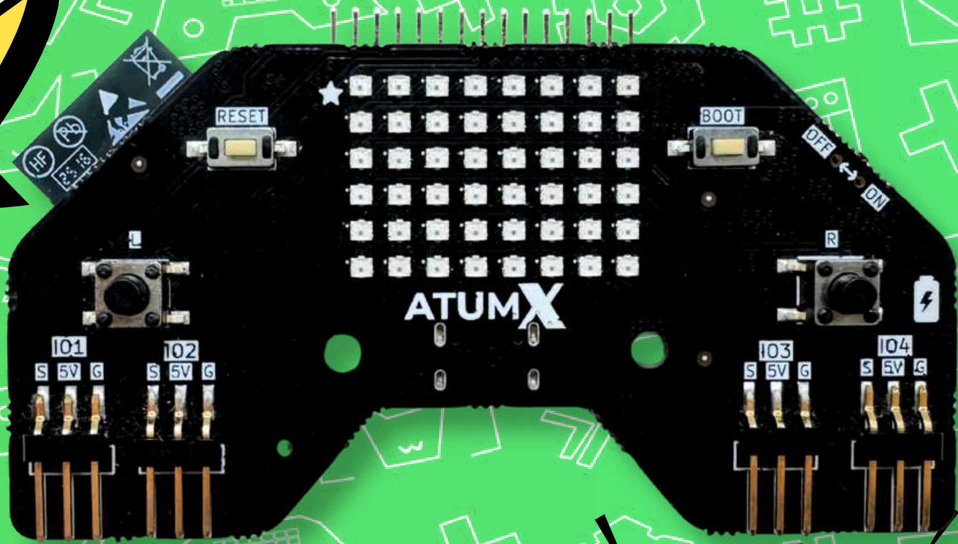


User manual and Secrets

Index

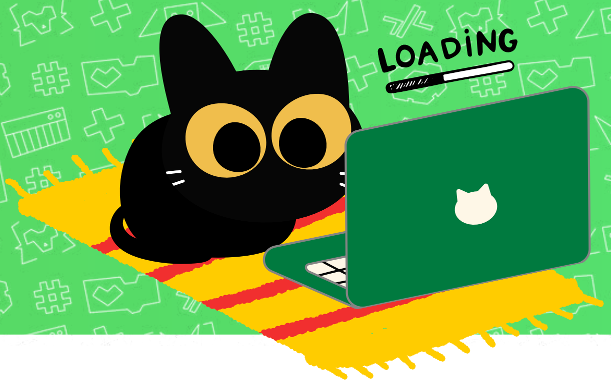
HI!!!



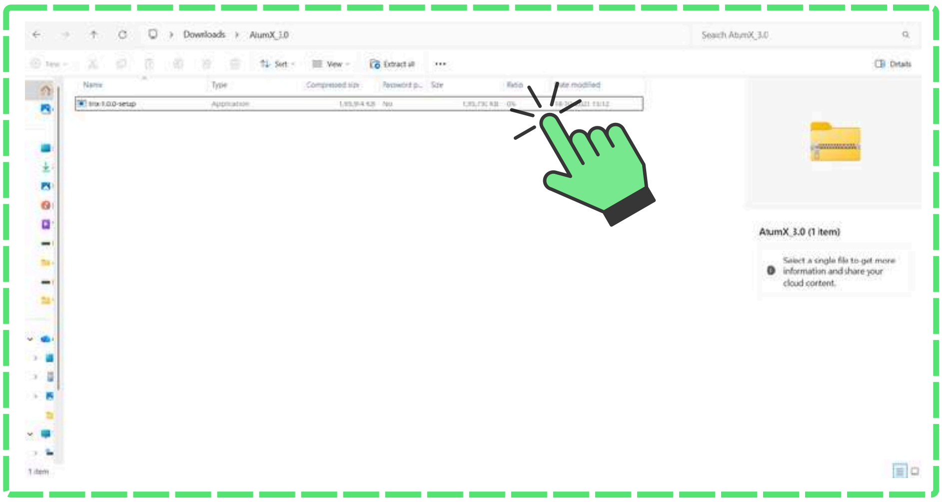
SUPPORTS MULTIPLE DIFFERENT EXPANSIONS

48	Jolly Lights
08	Pins of Power
01	Alarming Buzzer
02	Buttons

Infinite Possibilities

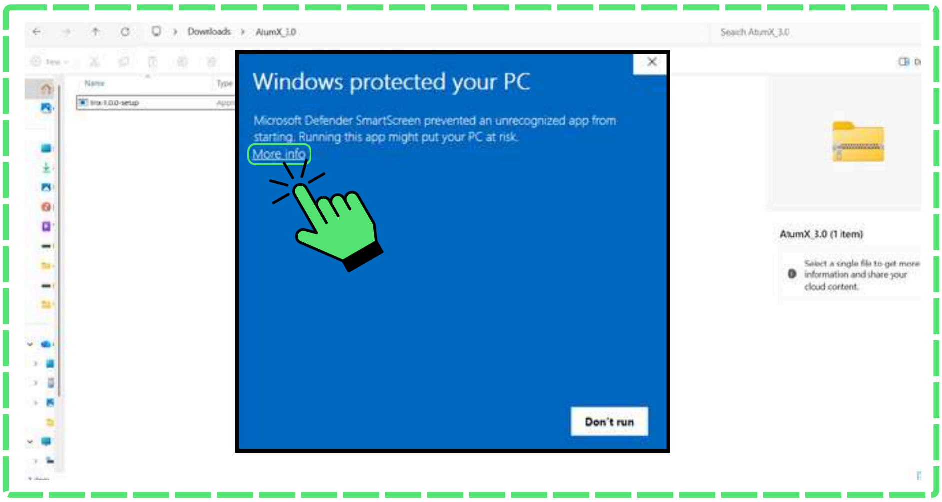


Installing the app



Step 1:

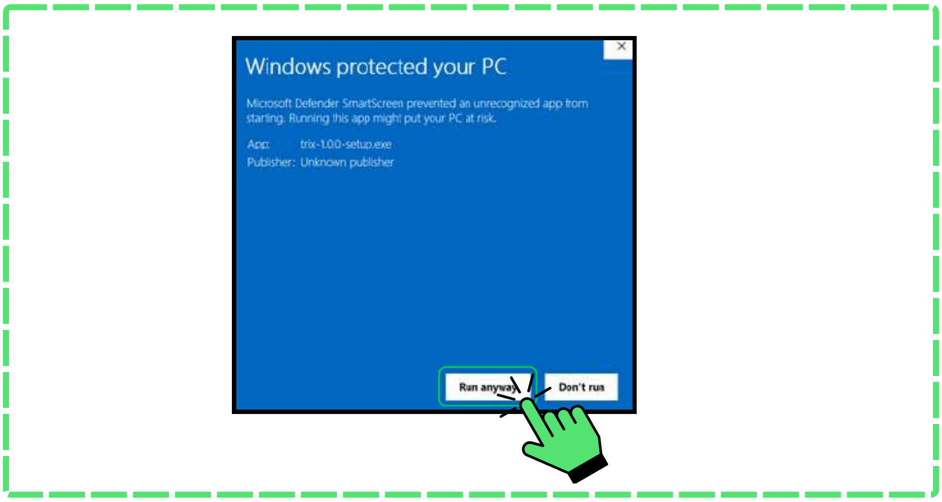
Click on the App you have downloaded from the QR code



Step 2:

A Windows pop up will open. **Don't panic, the App is safe.**

Click "More info" as shown here.



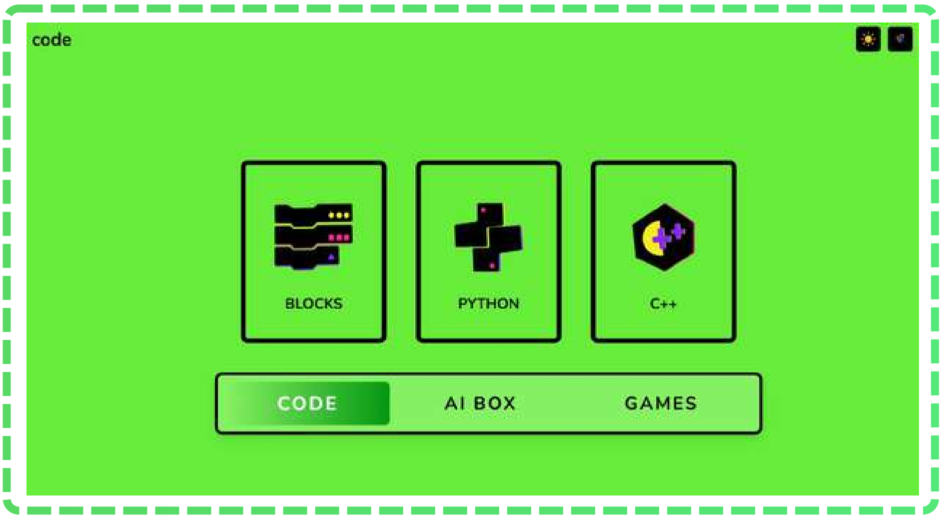
Step 3:

Click Run anyway and the app will be opened.

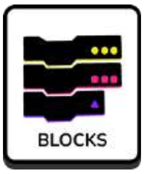
App features

CODE:

Home Screen



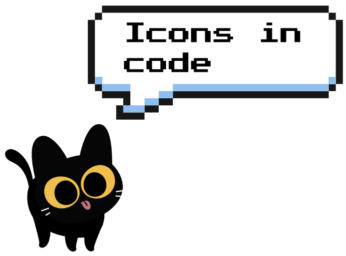
Blocks Mode



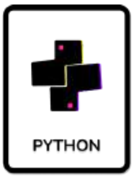
A visual drag-and-drop coding system using blocks.

Projects

Opens all the programs and activities you have created or saved.



Python



Text-based coding using the Python programming language.

Light/dark mode

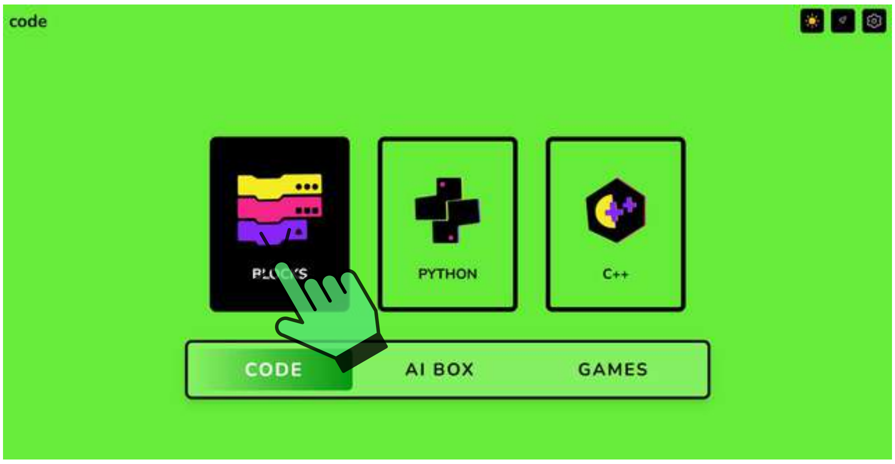
Switches the screen to light or dark colored theme.

C++

Coding in the C++ language for faster, performance-based programs.

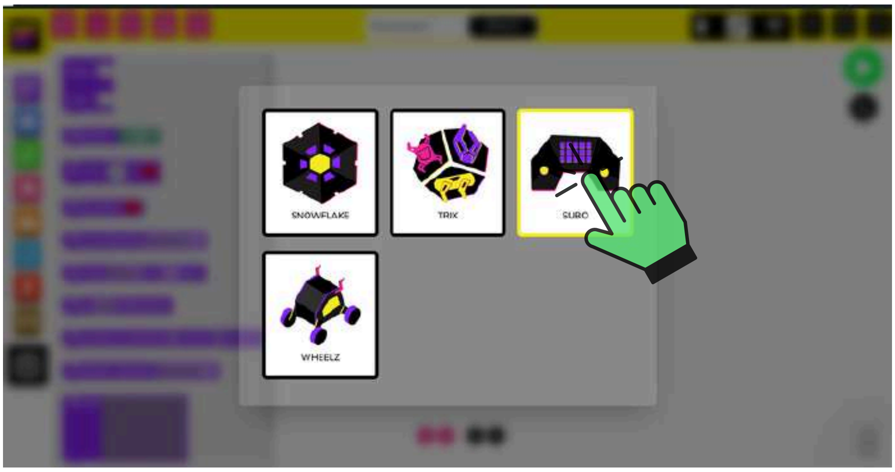
Blocks:

Entering blocks



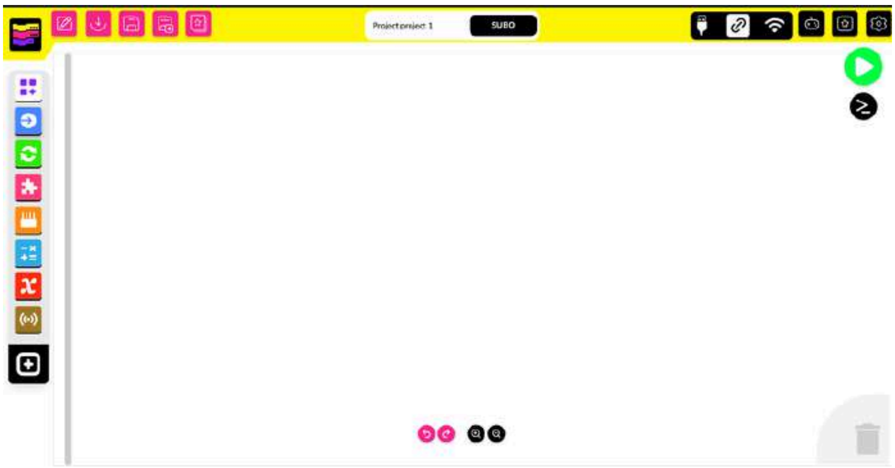
Step 1:

Click on the App you have downloaded and press Blocks in code section.



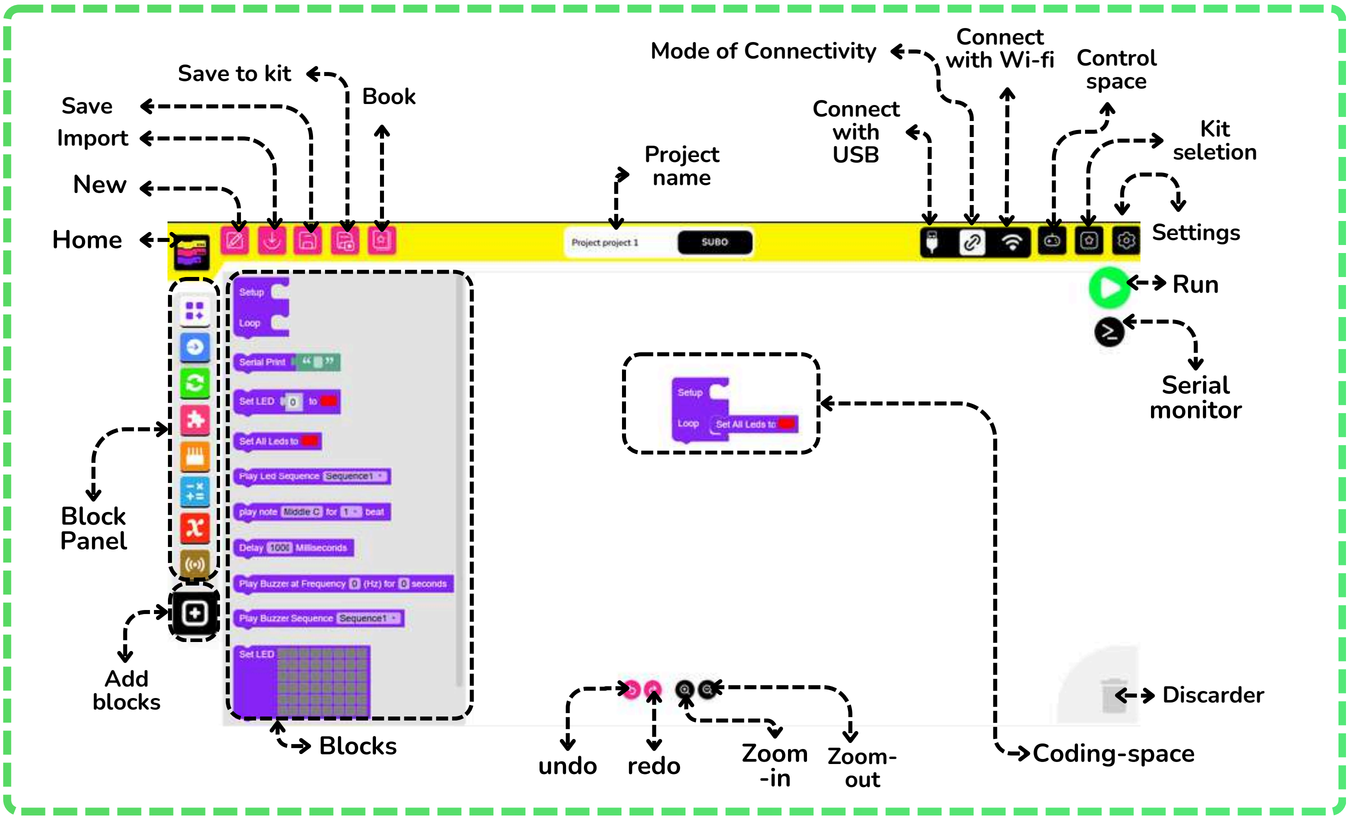
Step 2:

Click on Subo.



The app is now opened and ready for coding.

Parts



Blocks available:

- **BASIC** → Contains fundamental blocks like display text or pause.
- **ACTUATORS** → Contains various blocks to control actuators like motors, etc.
- **LOOP** → Repeats actions using loops like “repeat” or “forever.”
- **LOGIC** → Controls program flow using conditions like “if...then.”
- **I/O** → Manages input and output pins for external devices.
- **MATH** → Performs mathematical operations and calculations.
- **VARIABLE** → Stores and retrieves data using named variables.
- **SENSOR** → Reads data from various sensors like light or temperature.

Functions

Import
Loads a saved project.

Book
Opens learning resources or documentation.

Save to Kit
Uploads and stores the program directly to our hardware kit

Save
Saves the current project

New
Creates a new project

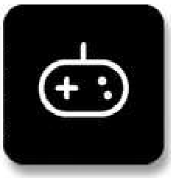
Home
Returns to the main screen.

Connect with USB
Links the device for uploading code using USB.

Connect with Wi-Fi
Links the device for uploading code using Wi-Fi.

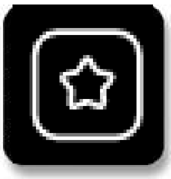
Mode of connectivity

Choose between Wi-Fi and USB mode of connecting our hardware



Control space

Has a dedicated remote control space for our hardware kits



Kit selection

Chooses the connected hardware kit.

Settings

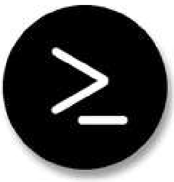
For app related settings.

Run

Executes the program.

Discarder

Removes unwanted blocks.



Serial monitor

Displays data from the connected device.

Coding-space

The area where the program is built using blocks.

Zoom in

Adjusts block view size.

Zoom out

Adjusts block view size.

Undo

Reverses the last change.

Redo

Restores the last change.

Blocks

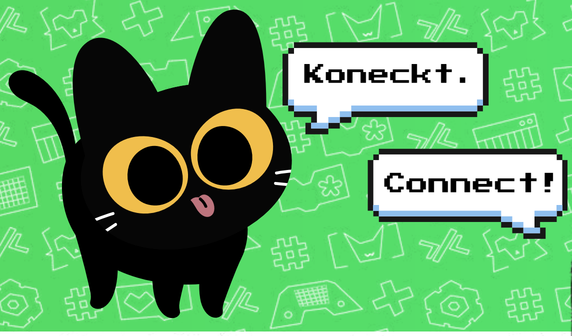
Displays all available block categories.

Block Panel

The sidebar containing categorized blocks.

Add blocks

For adding any other block extensions.



Connect with the app

Wired mode :

Step 1:

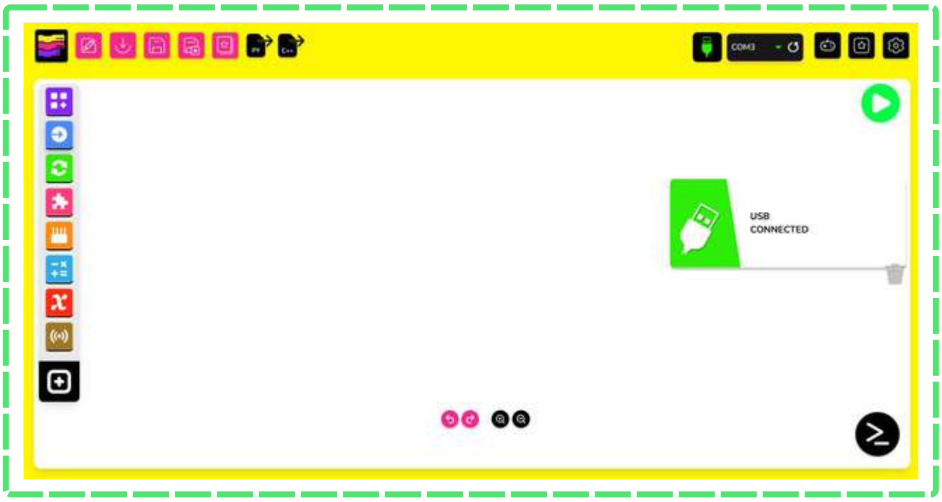
Click on the USB Connect icon .

Step 2:

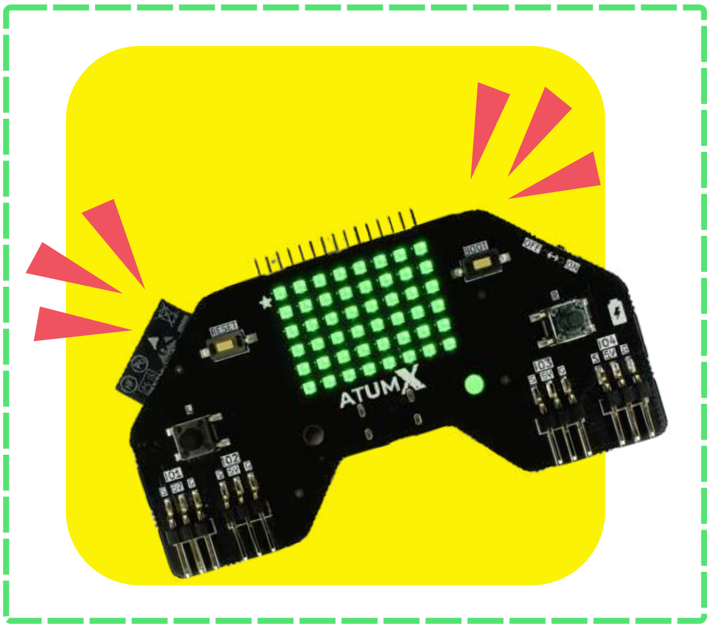
Click on the Drop-down box as shown.

Step 3:

Click on the COM port and wait for a popup.



Once the popup appears, the wired icon changes to green, & the board will glow with green LEDs



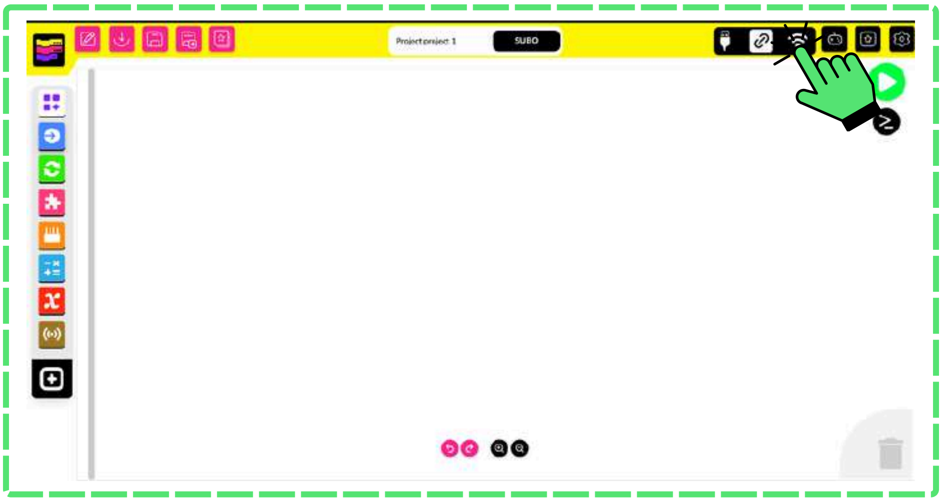
All the LEDs turn green, indicating that the board has been successfully connected.

Wireless mode :

Step 1:

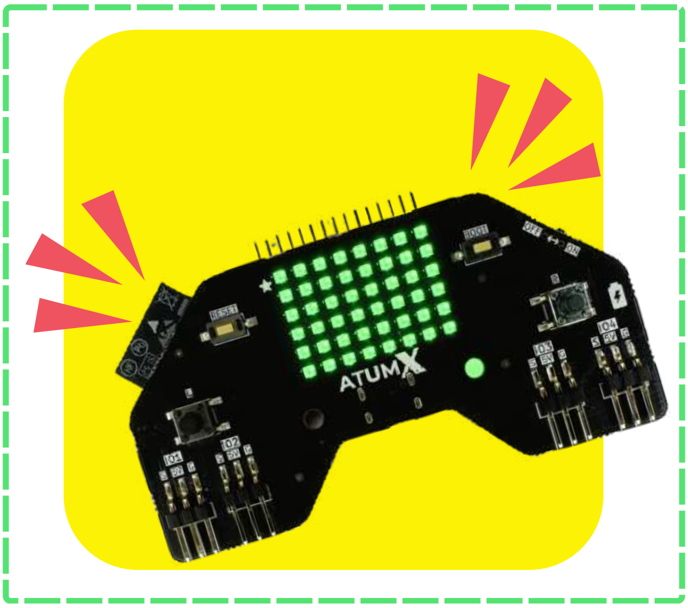
Open Wi-Fi Settings. Then Select "SUBU", Enter password (12345678) as shown and connect.

Step 2:



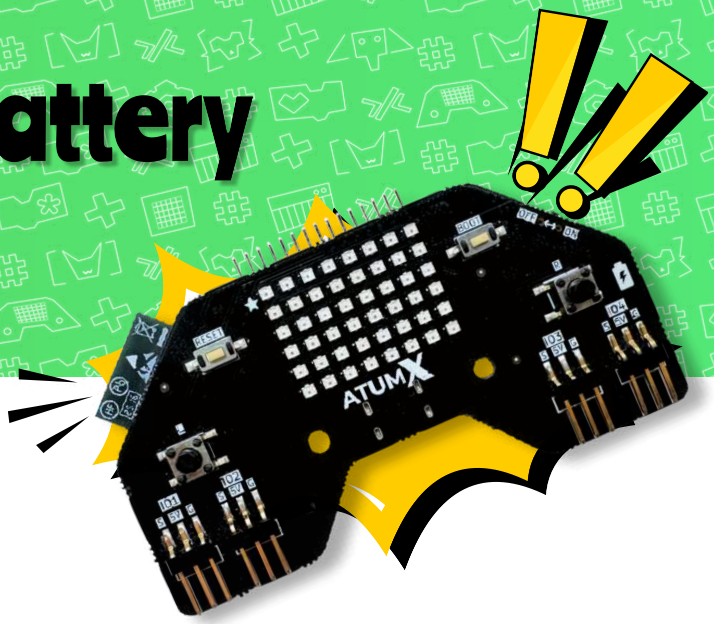
Click on the Wi-Fi icon as shown.

Once the popup appears, the Wi-Fi icon changes to green, & the board will glow with green LEDs



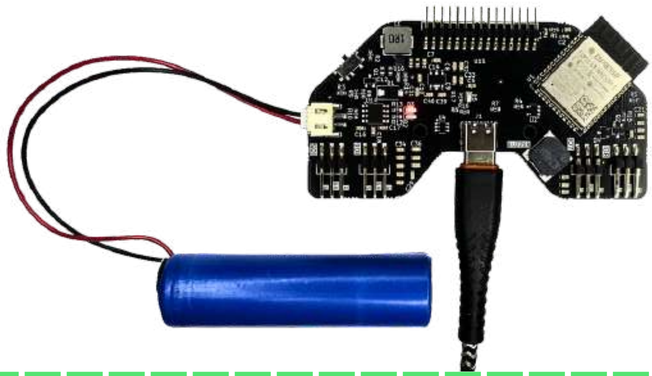
All the LEDs turn green, confirming that the board has been successfully connected "wirelessly".

Recharging our battery using our board!



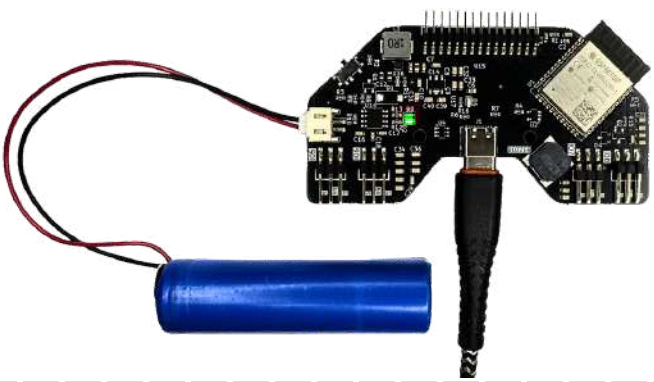
CAUTION

If you order the board by itself, the rechargeable battery isn't included - but don't worry, you can order one separately from us. The good news is that our Animal and Car kits already come with a battery included.



Charging :

LED glows RED = Battery is charging

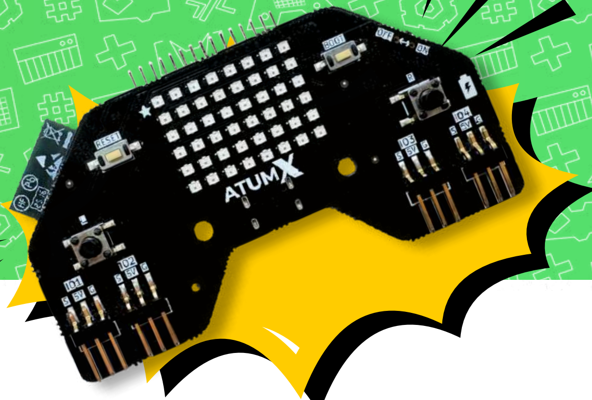


Battery full :

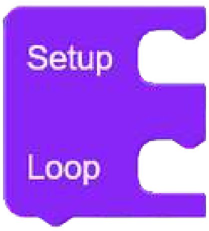
LED Glows GREEN =Battery is Full

OUR BATTERIES USUALLY TAKE 45MINS TO 2HRS TO FULLY RECHARGE

Board Intro



BEFORE YOU START: RUNS ONCE & RUNS CONTINUOUSLY!



Setup

The code blocks inside setup() run only once at the beginning. They're used to prepare the system like setting pin modes, starting sensors.

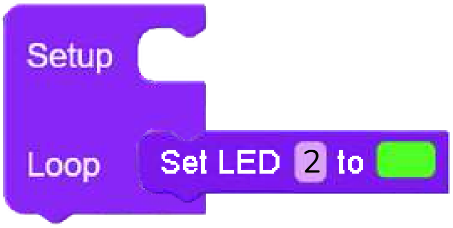
Loop

The blocks inside loop() run again and again in a continuous cycle. This is where the main program logic goes checking sensors, turning LEDs on/off, playing sounds, etc.

Control the LEDs

BLOCK PANEL	BLOCK	PURPOSE
	Set LED to	Turn on one led specified in the blank in particular color specified.
	Set All Leds to	Turn on all leds in particular color specified.

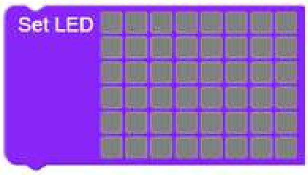
Examples



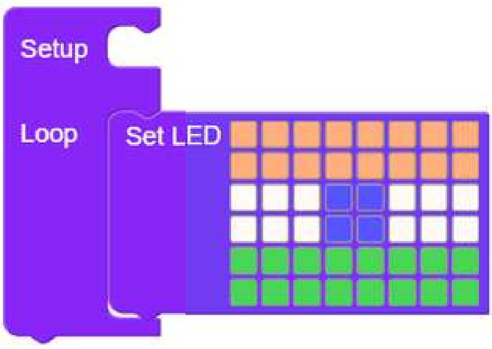
Turns on LED 2 in green color.

Turns on all LEDs in green color.

Create your own pattern!

BLOCK PANEL	BLOCK	PURPOSE
		Turn on LEDs selected in the blank in particular color specified.

Examples



The top two rows are made orange , center with white and the bottom two rows are green.

The LEDs glow as programmed!

Control the Buzzer

BLOCK PANEL	BLOCK	PURPOSE
	Play Buzzer at Frequency 0 (Hz) for 0 seconds	Play buzzer at particular frequency for specified time.
	play note Middle C for 1 beat 	Play buzzer at particular frequency for specified beat that you select.
	Play Buzzer Sequence Sequence1 (✓ Sequence1, Sequence2, Sequence3, Sequence4, Sequence5)	Play buzzer at particular sequence.

Examples

Setup | play note 349.23 for 1/2 beat
Loop

Play note middle F for 1/2 beat.

Setup | Play Buzzer at Frequency 500 (Hz) for 1 seconds
Loop

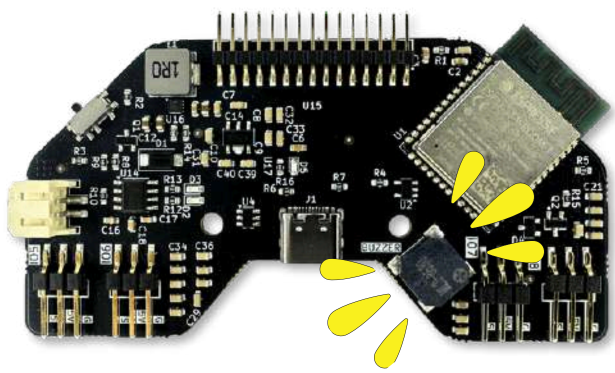
Play buzzer at 500 Hz for 1 second

Setup
Loop | Play Buzzer Sequence Sequence1

Play buzzer with sequence 1.

Add a **delay** when using **Play Note** or **Play Buzzer** blocks in **Loop**, and keep the **delay duration** equal to the **buzzer tone duration** as shown below

Setup
Loop | Play Buzzer at Frequency 500 (Hz) for 1 seconds
Delay 1000 Milliseconds

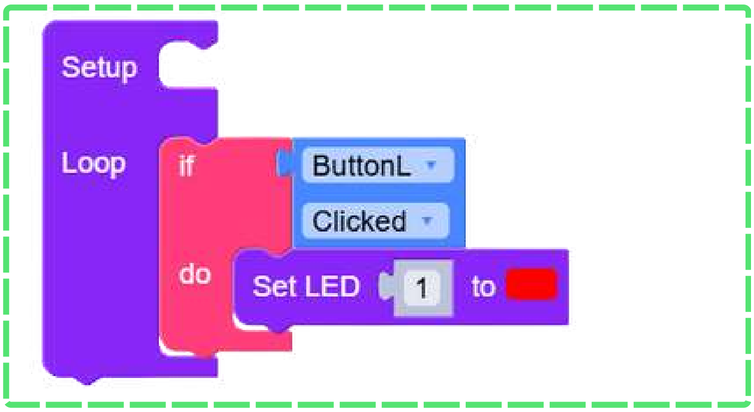


Voila! There goes your buzzer.

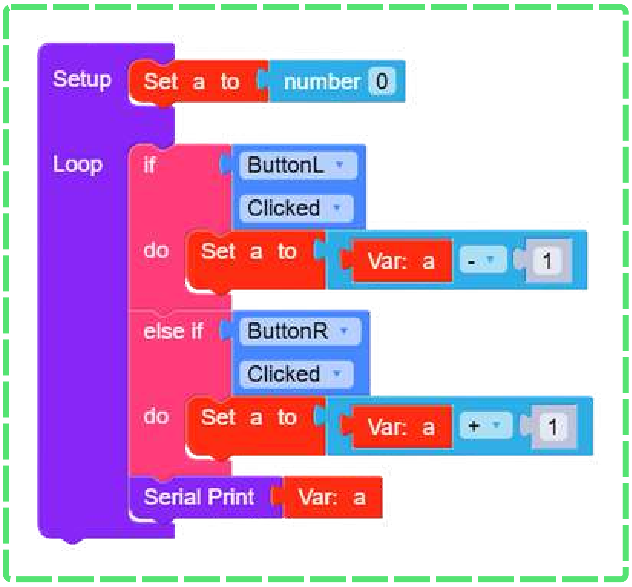
Control Buttons

BLOCK PANEL	BLOCK	PURPOSE
	ButtonL Clicked	Detects whether left button is clicked.
	ButtonR Clicked	Detects whether right button is clicked.

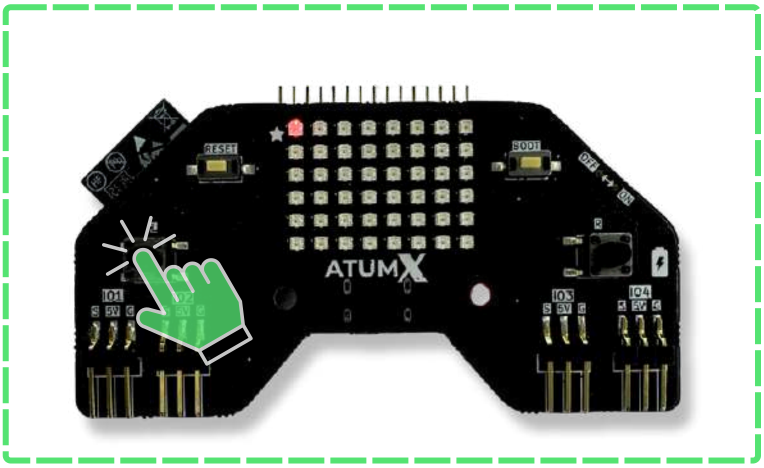
Examples



Turn on LED 1 in red color if left button is clicked.



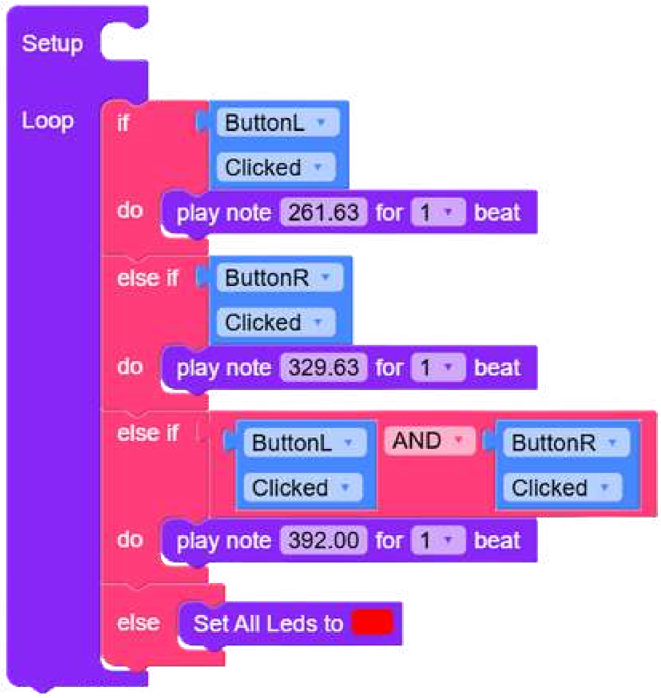
Setting variable values based on clicking left or right button.



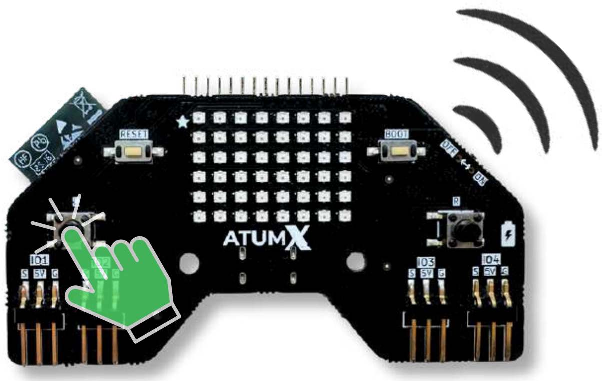
LED 1 glows red when left button is clicked.

Control Buttons

Examples



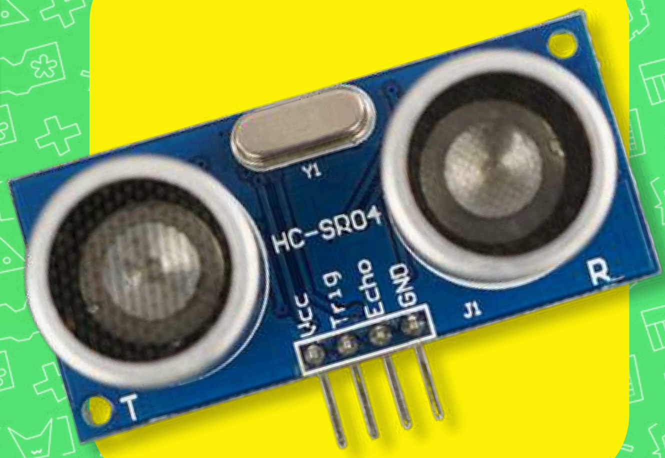
Play different notes for accessing left and right buttons and turn on all LEDs in red color if no buttons are pressed.



Plays a note when a button is pressed.

When no buttons are pressed.

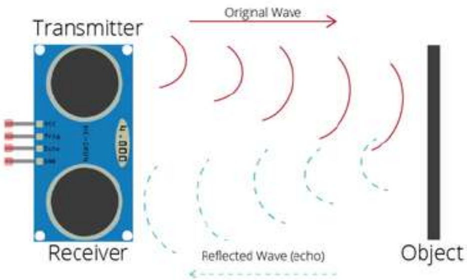
Ultrasonic sensor



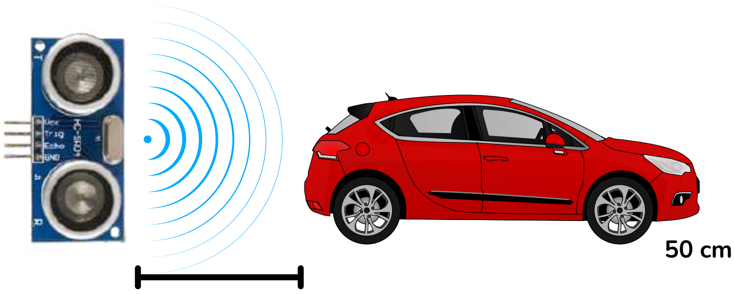
Know about ultrasonic sensor

Sensor sends out high pitched sound waves and waits for them to bounce back after hitting something. By measuring how long the echo takes to return, it can figure out how far away the object is. The sensor emits ultrasonic waves (usually around 40 kHz), which travel through the air, reflect off an object, and return to the sensor's receiver.

- The distance is calculated using the formula: $\text{Distance} = \dfrac{\text{Speed of Sound} \times \text{Time}}{2}$
- The division by 2 is because the sound has to travel to the object and back.
- It has two main parts:
    a. **Transmitter** (TX): Sends out the ultrasonic pulse.
    b. **Receiver** (RX): Listens for the echo.



Sensor Working

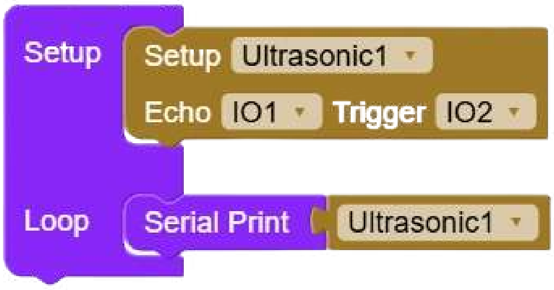


The ultrasonic sensor emits waves that bounce off an object. By measuring the return time, it calculates the distance. As the object moves farther, the return time increases, giving a higher distance reading.

Models Compatible

HC-SR04

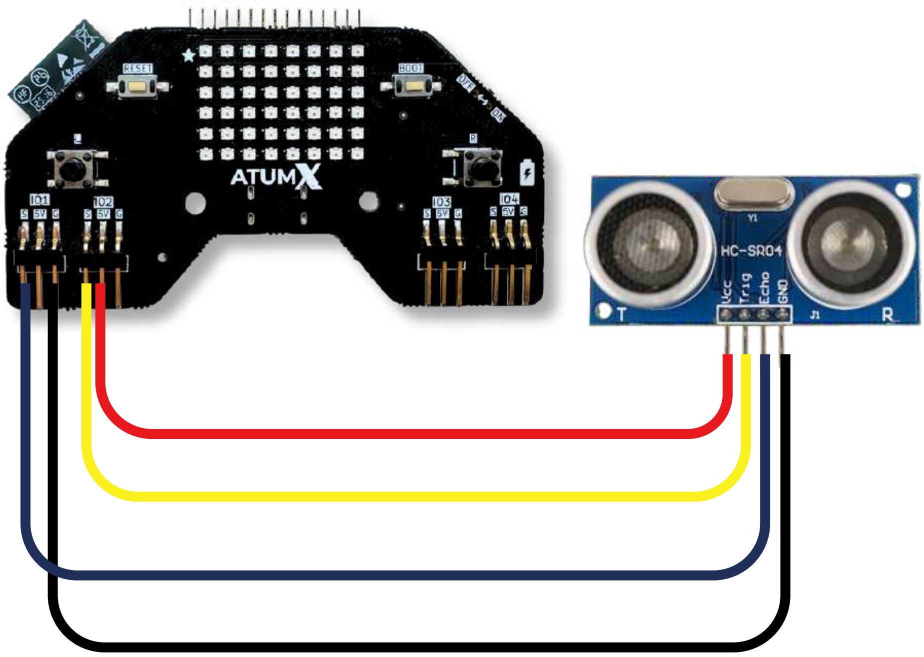
Block Code



Pin Connection

Board	Sensor
5V	VCC
Gnd	Gnd
Connect to **S(IO1)** or any other **available IO pin** defined in the snippet block.	ECHO
Connect to **S(IO2)** or any other **available IO pin** defined in the snippet block.	TRIG

Video link on how to connect



Power line

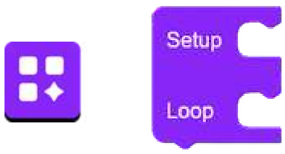
Trigger

Echo

Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Configure the ultrasonic sensor pins. Select any **available GPIO** for **Trigger** and **Echo** using this block

Block 3:

This block **reads the distance** measured by the selected ultrasonic sensor

Block 4:

Use **Serial Print block** with ultrasonic block to display Ultrasonic sensor values on the **Serial Monitor.**

DID YOU KNOW?

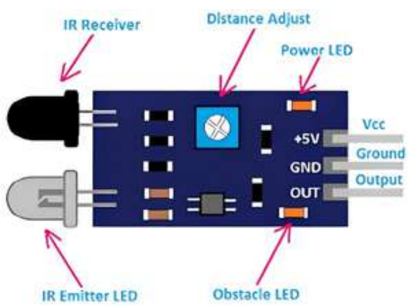
Ultrasonic sensors measure distance by sending out sound waves and calculating how long it takes for them to bounce back.

IR sensor

Know about IR sensor

An IR (Infrared) sensor uses infrared light to detect objects or measure distance. It works by emitting IR light from an IR LED and checking how much of it is reflected back by an object using a photodiode.

- If more light is reflected, the object is closer.
- If little or no light is reflected, the object is far or absent.



It has two main parts:

a. Emitter (IR LED): Sends out infrared light.

b. Receiver (Photodiode): Detects the reflected IR light.

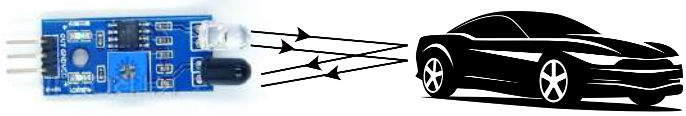
IR sensors are widely used for obstacle detection, proximity sensing, and line-following robot

Sensor Working

a)

a) No object present – IR rays are not reflected back, so no detection.

b)



b) Object present – IR rays reflect back to the receiver, object detected.

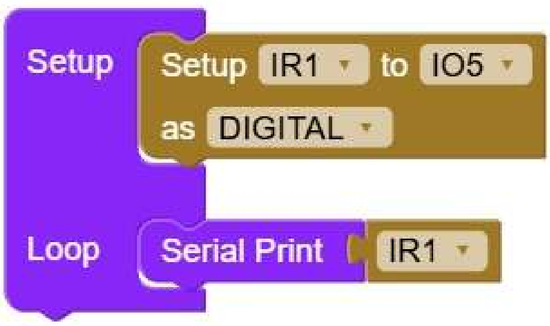
Models Compatible

3 pins- LM393
4 pins- TCRT5000

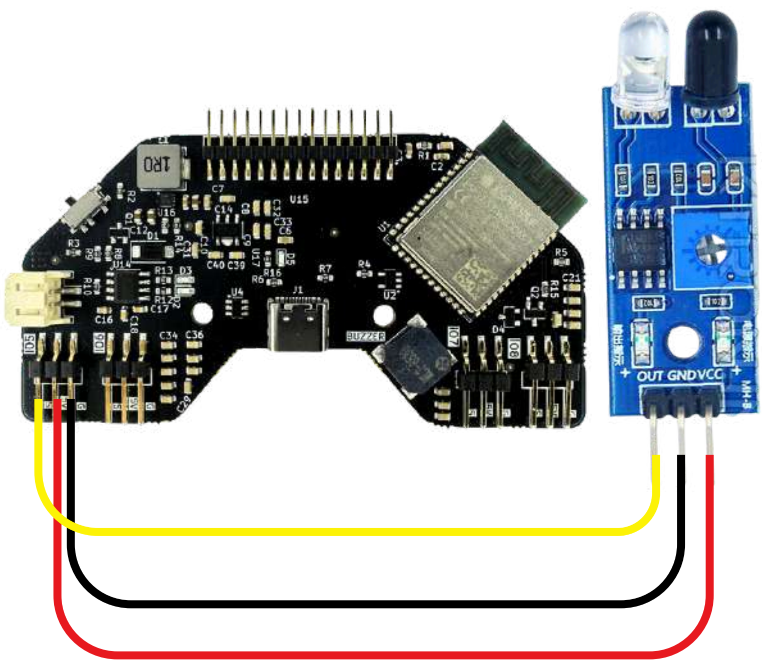
Pin Connection

Board	Sensor
5 V	VCC
G	GND
Select S(IO5) or any other available IO pin defined in the snippet block.	AO/DO

Block Code



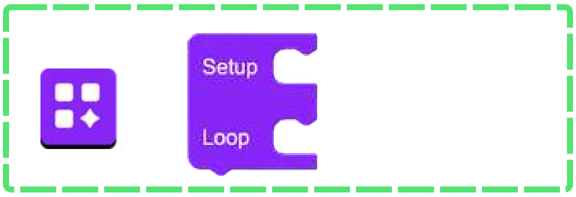
Video link on how to connect



Power line
DO
Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.

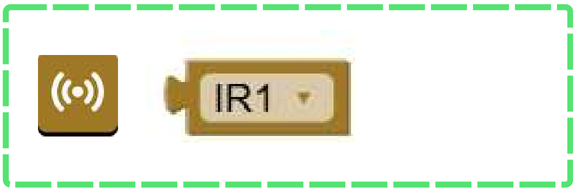


Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Configure the IR sensor pin. Select any available GPIO code snippet and set it as DIGITAL using this block.



Block 3:

This block reads the object detection status (object present or not) from the IR sensor.

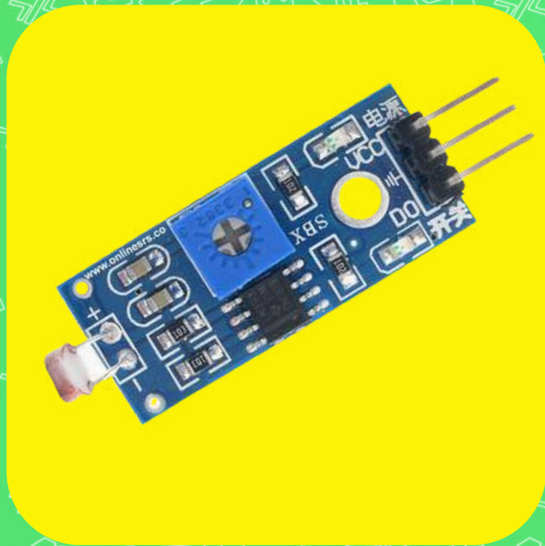
Block 4:

Use **Serial Print block** with IR block to display IR sensor values on the **Serial Monitor.**

DID YOU KNOW?

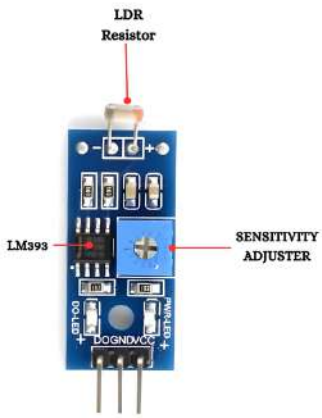
IR sensors can "see" heat! They detect infrared radiation, which is emitted by all warm objects—even your hand!

LDR sensor



Know about LDR sensor

An LDR (Light Dependent Resistor) is a sensor whose resistance changes based on the amount of light falling on it. In bright light, its resistance becomes very low, allowing more current to pass. In darkness, its resistance becomes very high, restricting current flow. By measuring this change in resistance, a microcontroller can determine the light level .



Working Principle:
The sensor works on the principle of photoconductivity the conductivity of certain materials increases when they are exposed to light..

In Fig.1 When light falls on the LDR, its resistance decreases, allowing more current to flow.
In Fig.2 In darkness, the LDR's resistance increases, reducing the current flow.

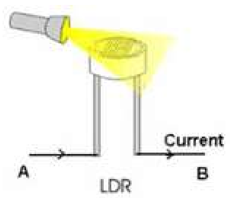


Fig.1

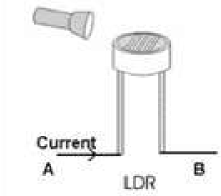
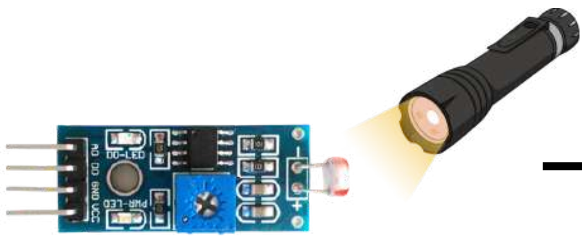


Fig.2

Sensor Working

a)



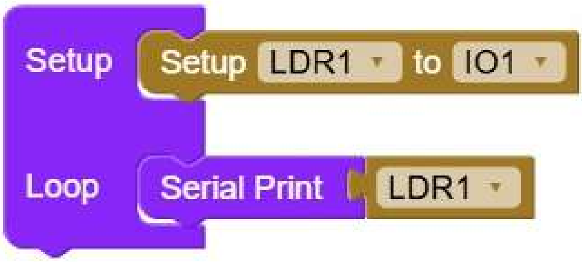
When exposed to light, the LDR output decreases (e.g., closer to 0) b)

When exposed to darkness, the output increases (e.g., closer to 4095).

Models Compatible

LM393 based

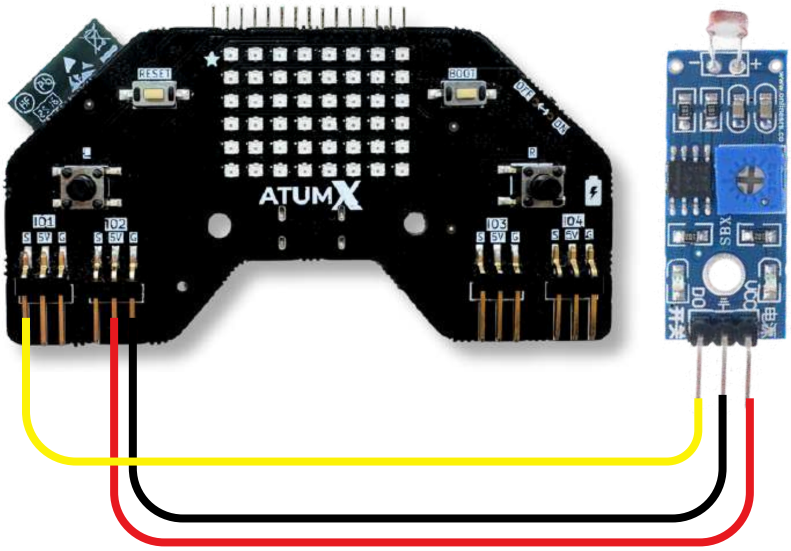
Block Code



Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO1 or any **available GPIO Pin** defined in the snippet block .	AO/DO

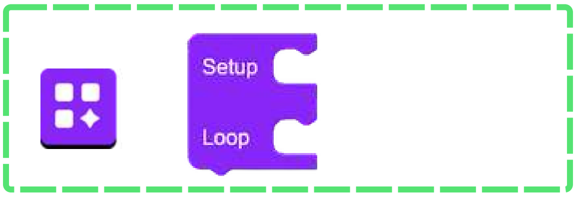
Video link on how to connect



Power line
DO
Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Setup LDR** block to assign the LDR sensor to **any GPIO pin** on the board.

Block 3:

Use the **LDR block** to read the light intensity detected by the Light Dependent Resistor

Block 4:

Use **Serial Print** block attached with LDR block to print LDR sensor values on the **Serial Monitor.**

DID YOU KNOW?

LDR sensors are used in streetlights to automatically turn on at night and off during the day by detecting changes in ambient light levels.

Gas Sensor

Know about gas sensor

A Gas Sensor detects the presence and concentration of gases in the air. It works by changing its electrical resistance when it comes in contact with certain gases. The sensor has a sensing element (usually made of metal oxide like SnO_2) that reacts chemically with gas molecules. This reaction changes the current flowing through the sensor.

It has two main parts:

a. Sensing Element: Detects the gas by reacting with its molecules.

b. Circuit Module: Converts the change in resistance into an electrical signal that can be measured

Working principle:

- When no gas – resistance of the sensing material remains normal.
- When gas is present – the gas molecules react with the sensing surface, changing its resistance.
- The circuit detects this change and sends a corresponding output signal.

Sensor Working

When gas is present, sensor reacts and produces an electrical signal.

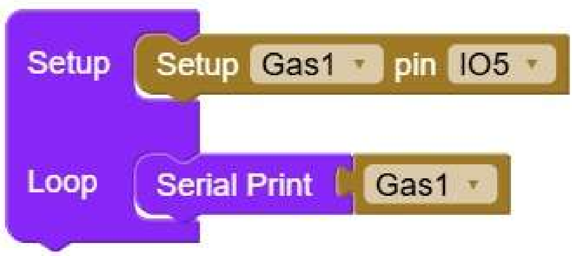
Models Compatible

MQ-2

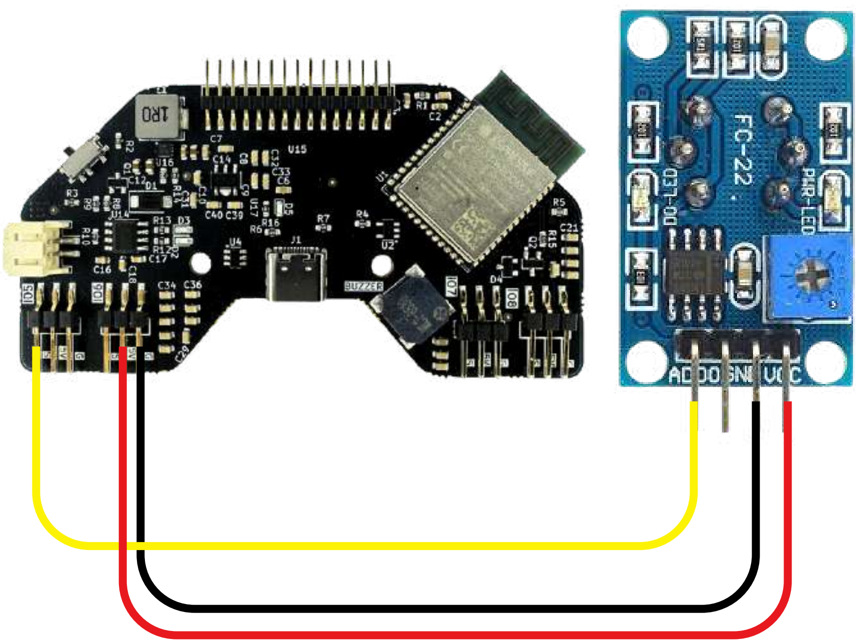
Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO5 or any **available GPIO Pin** defined in the snippet block .	AO/DO

Block Code



Video link on how to connect



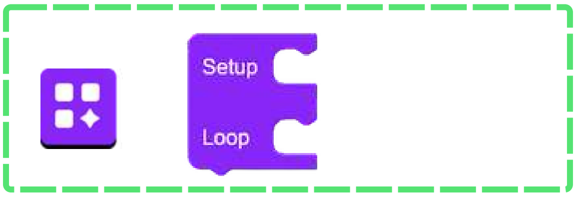
Power line

DO

Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

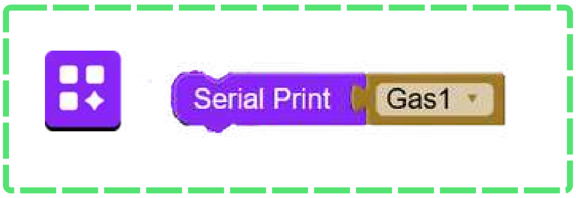
From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Set Gas** block to assign the gas sensor to **any GPIO pin** on the board.

Block 3:

Use the **Gas block** to read the output signal



Block 4:

Use **Serial Print** block attached with gas block to print gas sensor values on the **Serial Monitor.**

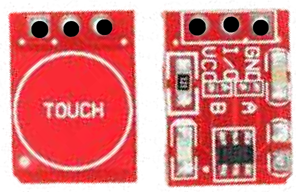
DID YOU KNOW?

Gas sensors are sensitive not only to the target gas but also to temperature and humidity, which can slightly affect their readings.

Touch Sensor

Know about Touch sensor

A **Touch Sensor** detects physical contact or touch on its surface. It works by sensing a small change in capacitance or resistance when a human finger touches it. When you touch the sensor, it generates an electrical signal that can be used to trigger an output device like an LED or buzzer.

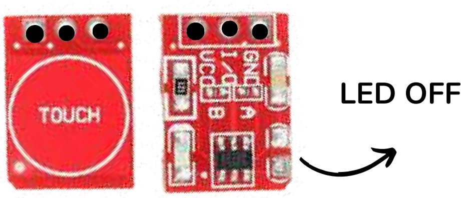


Working principle:

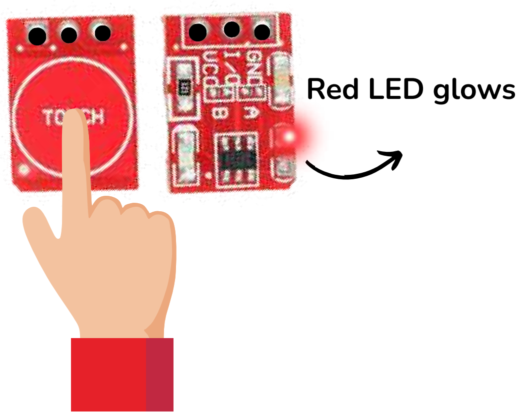
- When no touch – the circuit remains in its normal state.
- When touched – the human body (which conducts electricity slightly) changes the capacitance, and the circuit detects this change as a "touch."

Sensor Working

When no touch,LED does not glow.



When touched,LED glows.



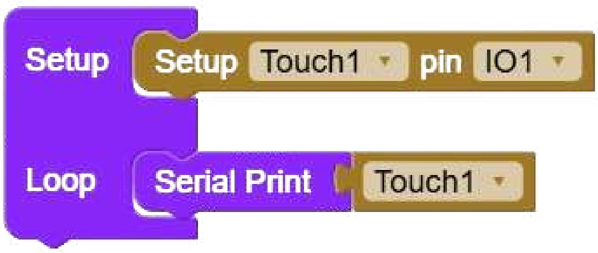
Models Compatible

TTP223B : 1 Channel (blue)
TTP223R : 1 Channel (red)

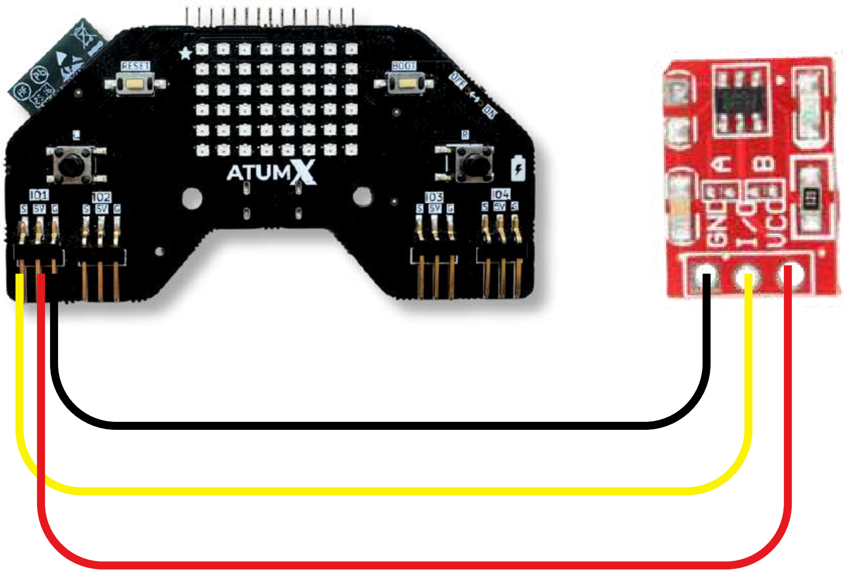
Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO1 or any **available GPIO Pin** defined in the snippet block .	AO/DO

Block Code



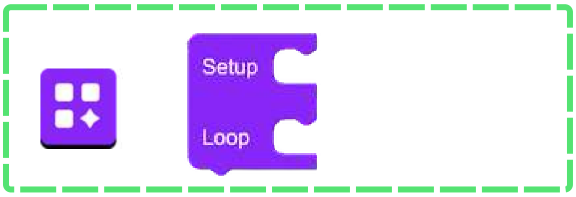
Video link on how to connect



Power line
DO
Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

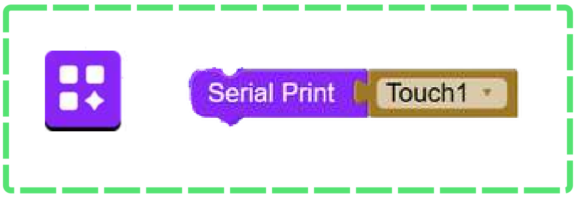
From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Set touch pin** block to assign the touch sensor to **any GPIO pin** on the board.

Block 3:

Use the **Touch block** to read the output signal



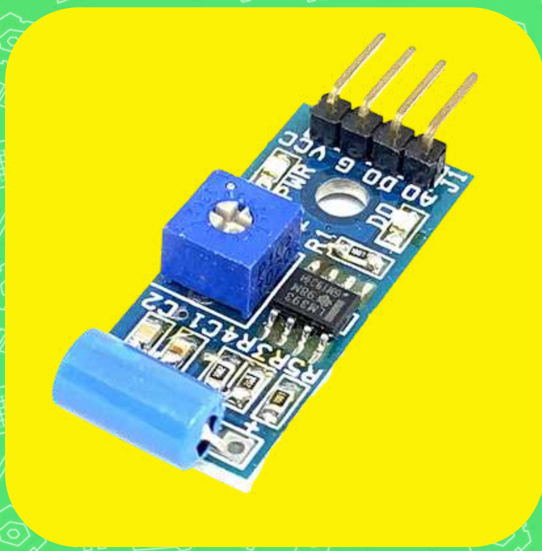
Block 4:

Use **Serial Print** block attached with Touch block to print Touch sensor values on the **Serial Monitor.**

DID YOU KNOW?

Touch sensors are used in mobile screens, elevator buttons, automatic doors, and smart home systems. They make devices more interactive and user-friendly.

Vibration Sensor



Know about Vibration sensor

A Vibration Sensor detects vibrations, shocks, or movements from surfaces or objects.It converts mechanical vibration into an electrical signal that can be measured by a microcontroller or circuit.It is often used to detect motion, impacts, or mechanical faults in machines.

Main Parts:

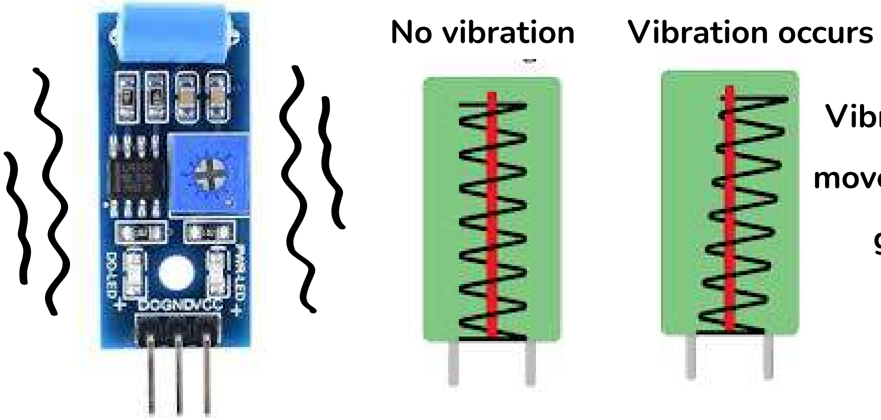
a. Sensing Element (Spring or Piezoelectric Material): Detects vibration or movement.

b. Signal Conditioning Circuit: Converts the vibration into a readable electrical signal.

Working Principle:

- When no vibration – the sensor remains in its normal state and output stays LOW.
- When vibration occurs – mechanical movement causes a change in voltage (in a piezoelectric type) or closes a contact (in a spring type).
- The circuit detects this change and sends an output signal (HIGH) to indicate vibration.

Sensor Working

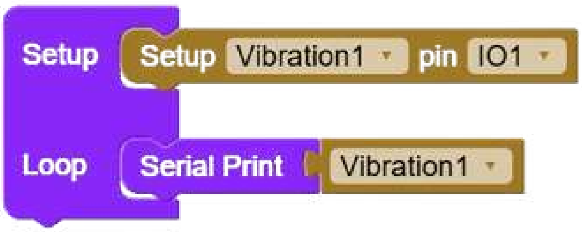


Vibration causes spring to move and close the contact, generating a signal.

Models Compatible

SW-420

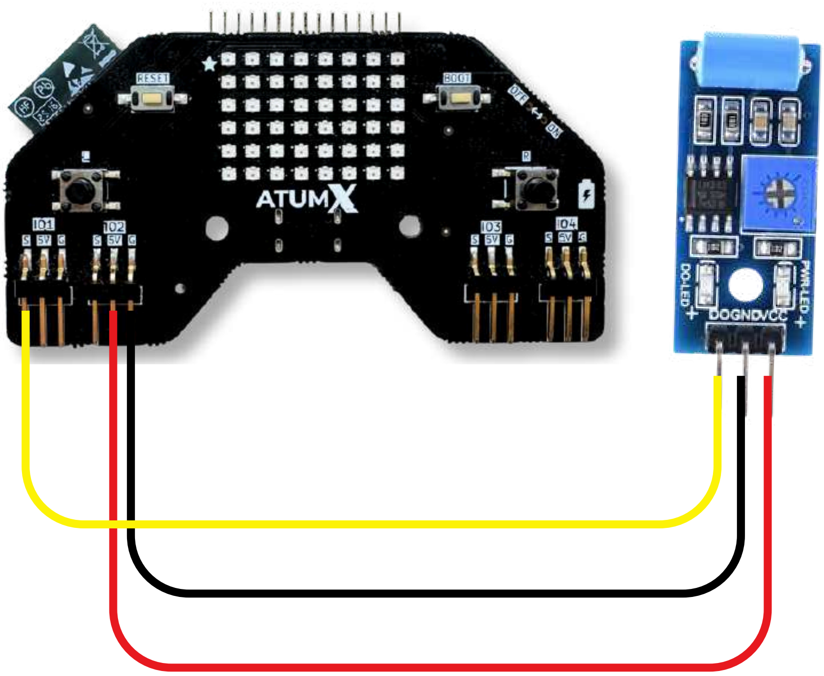
Block Code



Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO16 or any **available GPIO Pin** defined in the snippet block .	AO/DO

Video link on how to connect



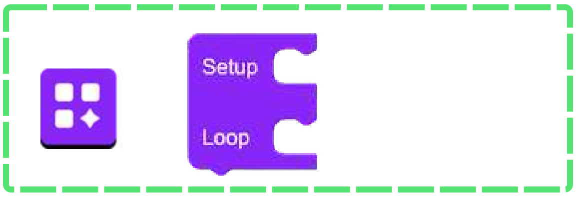
Power line

DO

Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Setup Vibration pin** block to assign the vibration sensor to any GPIO pin on the board.

Block 3:

Use the **Vibration block** to read the output signal

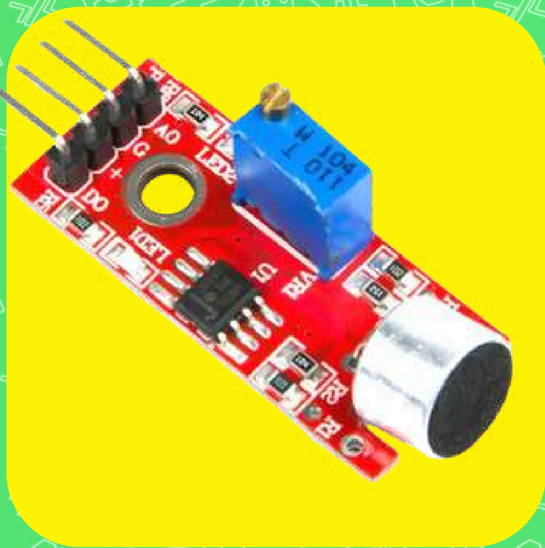
Block 4:

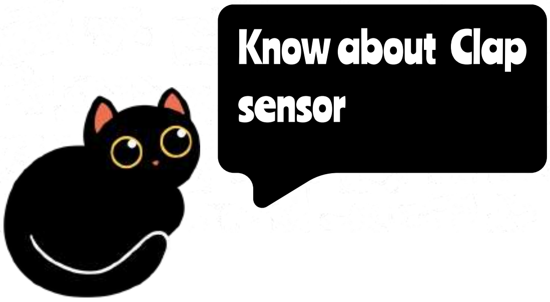
Use **Serial Print** block attached with Vibration block to print vibration sensor values on the **Serial Monitor.**

DID YOU KNOW?

A vibration sensor can detect even tiny movements that humans cannot feel, which is why it's used in machines, alarms, and earthquake-monitoring systems.

Sound Sensor





A Sound Sensor is used to detect sound vibrations, especially sharp sounds like claps, and convert them into electrical signals that can be read by a microcontroller or used to trigger circuits (like turning ON lights, motors, etc.).

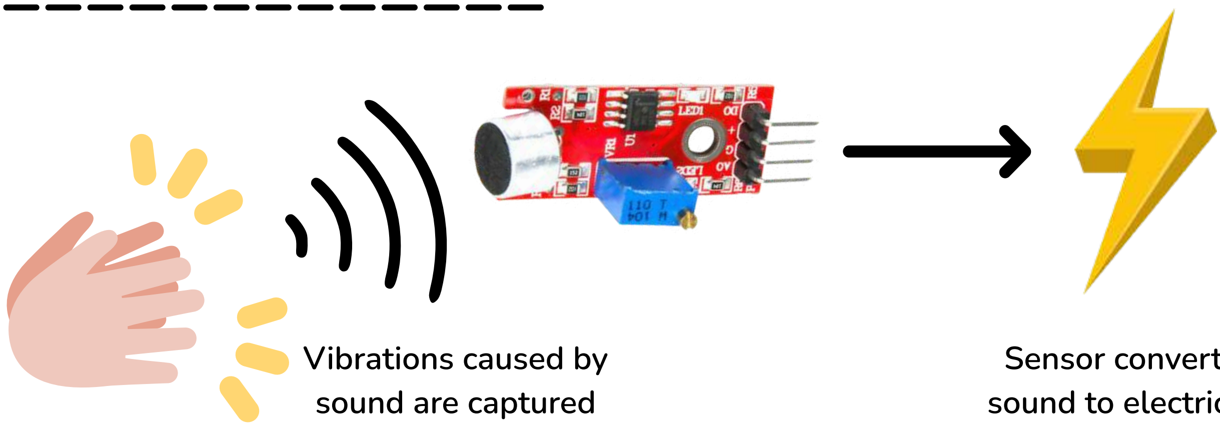
Main Parts

- Microphone: Detects sound waves and converts them into weak electrical signals.
- Amplifier: Strengthens the weak audio signals.
- Signal Processing Unit: Checks if the sound crosses a set threshold and outputs a clean ON/OFF pulse.

Working Principle:

- A sound creates vibrations that the microphone converts into a small signal.
- The amplifier boosts this signal, and the processing unit compares it to a threshold.
- If the sound is loud and sharp enough, the circuit outputs a digital HIGH pulse to trigger devices like LEDs or relays.

Sensor Working



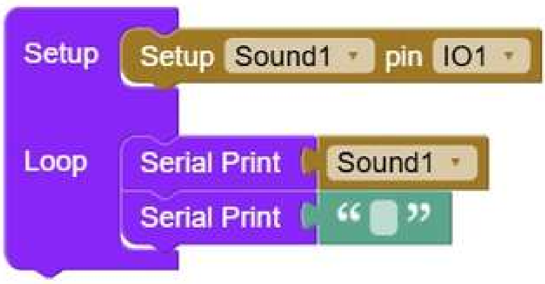
Vibrations caused by sound are captured by microphone

Sensor converts sound to electrical energy

Models Compatible

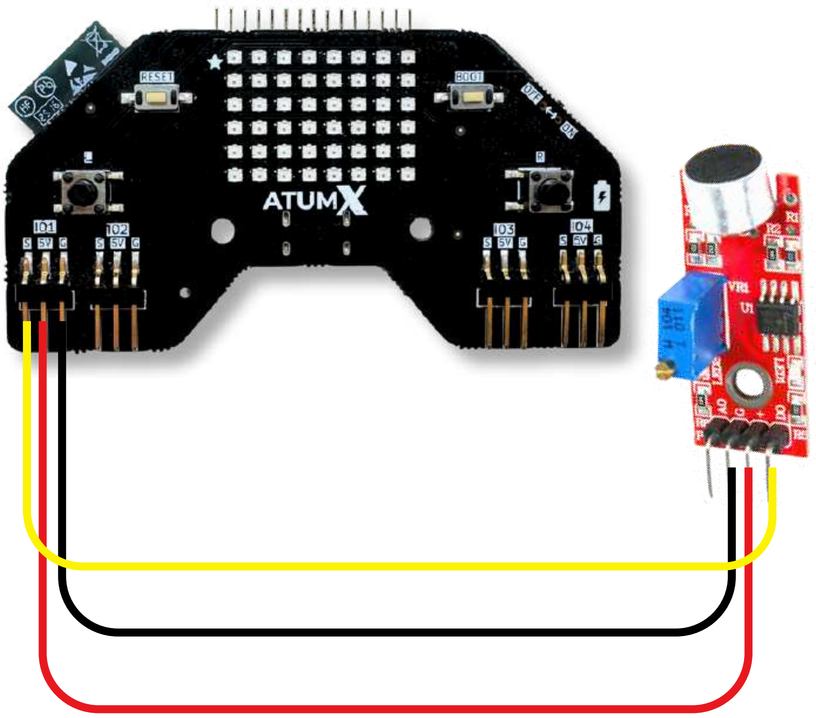
LM393 op-amp

Block Code



Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO1 or any **available GPIO Pin** defined in the snippet block .	AO/DO



Video link on how to connect

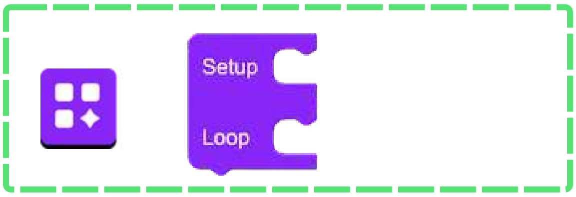
Power line

DO

Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

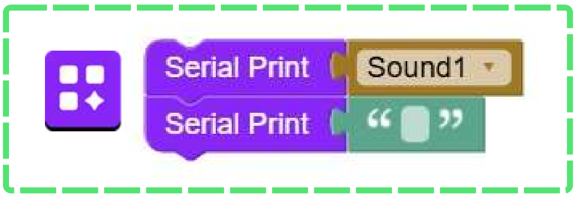
From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Set Sound** block to assign the sound sensor to **any GPIO pin** on the board.

Block 3:

Use the **Sound block** to read the output signal



Block 4:

Use **Serial Print** block attached with sound block to print sound sensor values on the **Serial Monitor.**

DID YOU KNOW?

A sound sensor's output depends on both sound intensity and distance — the farther the sound source, the weaker the response.

DHT11

Know about DHT sensor

The DHT11 Sensor is used to measure temperature and humidity in the surrounding environment. It provides digital output signals that can be directly read by a microcontroller . The sensor contains both a humidity sensing component and a temperature measuring component inside a single module.

Main Parts:

a. Humidity Sensor: Measures the moisture content in the air.

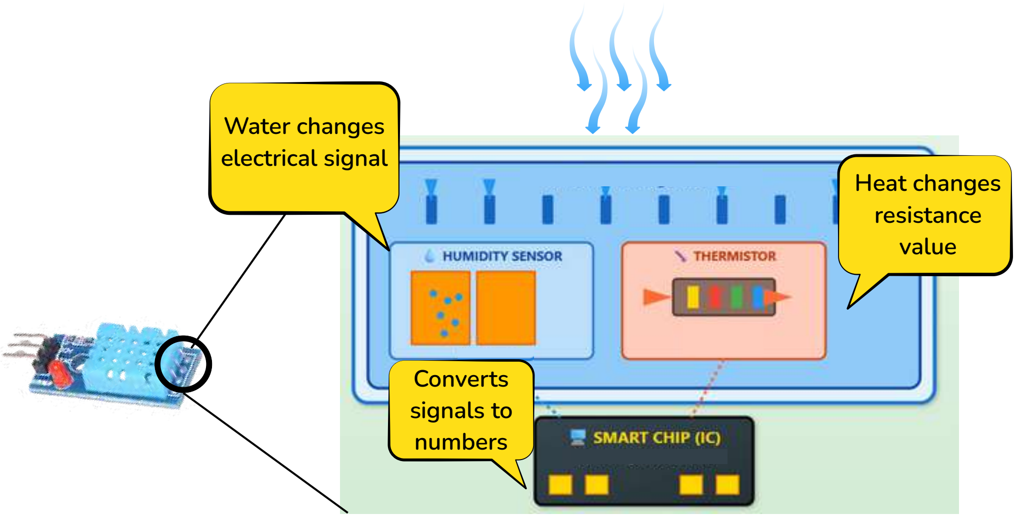
b. Temperature Sensor (Thermistor): Measures the surrounding temperature.

c. Signal Processing Unit: Converts the readings into digital data for easy interfacing.

Working Principle:

- The humidity sensor uses a moisture-holding substrate that changes electrical resistance with humidity.
- The thermistor changes resistance with temperature.
- The sensor processes both signals and sends the combined data as a digital output.

Sensor Working

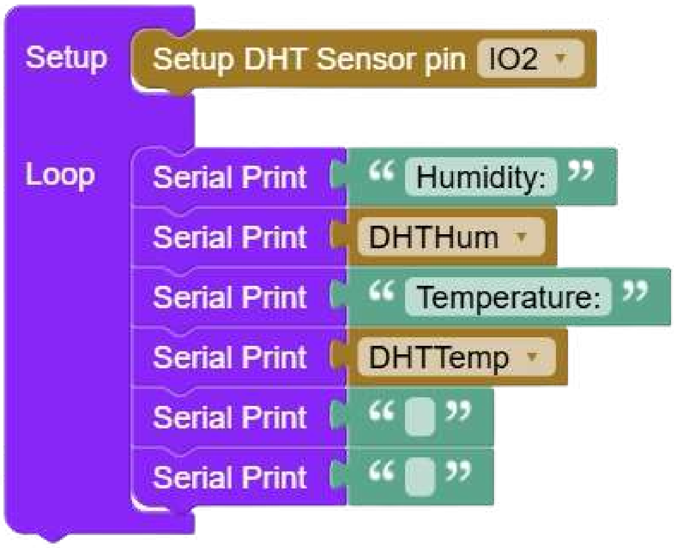


A DHT11 works by sensing air moisture through a humidity-sensitive material and temperature through an internal resistor, then sending the data digitally to a microcontroller.

Models Compatible

DHT11

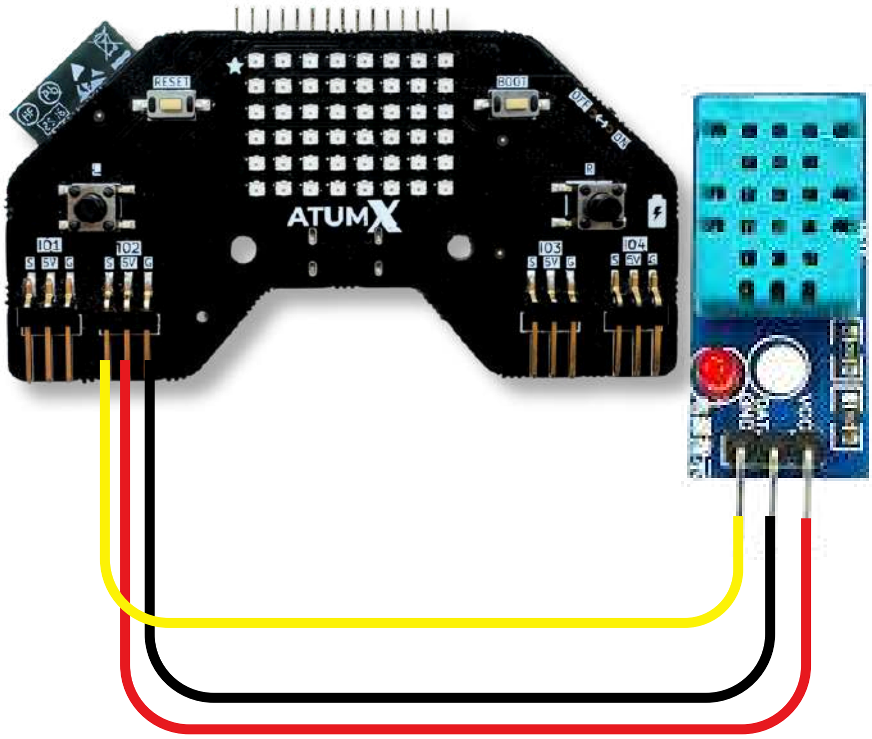
Block Code



Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO2 or any **available GPIO Pin** defined in the snippet block .	DATA

Video link on how to connect



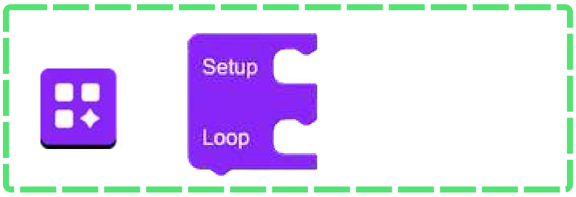
Power line

DO

Ground line

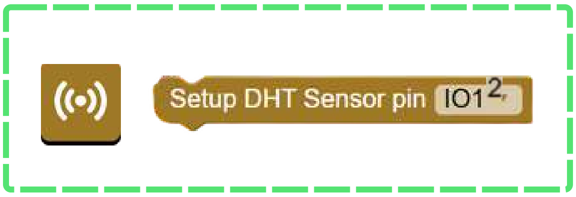
Check the IO pins you connected in the board and in the code snippet are same.

Code explanation



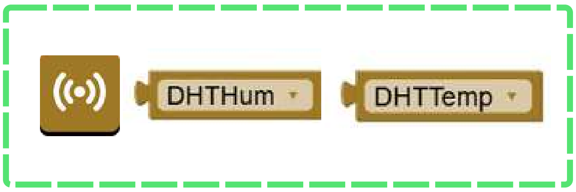
Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**



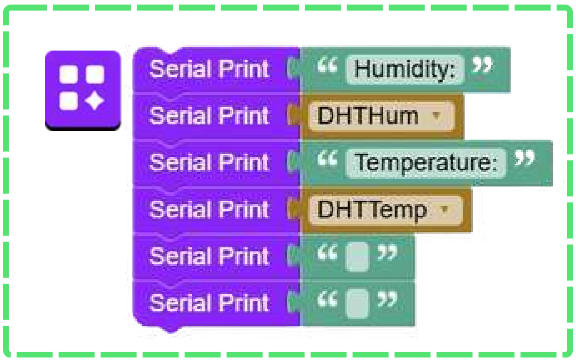
Block 2:

Use the **Setup DHT Sensor pin** block to assign the DHT sensor to **any GPIO pin** on the board.



Block 3:

Use the **Humidity and Temperature blocks of DHT** to read the output signal



Block 4:

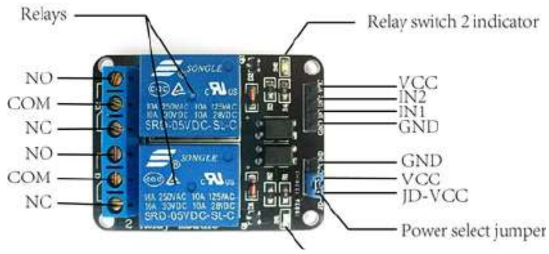
Use **Serial Print** block attached with DHTHum and DHTTemp blocks to print DHT sensor values on the **Serial Monitor.**

DID YOU KNOW?

Although it looks basic, the DHT11 actually sends data in a 40-bit digital packet, including humidity, temperature, and a checksum for error checking.

Relay Module - 2 Relay

Know about Relay module



A 2-Relay Module lets you control two high-power devices (like bulbs or motors) using a small control signal from Arduino, ESP32, or Raspberry Pi. It works like an electronic ON/OFF switch.

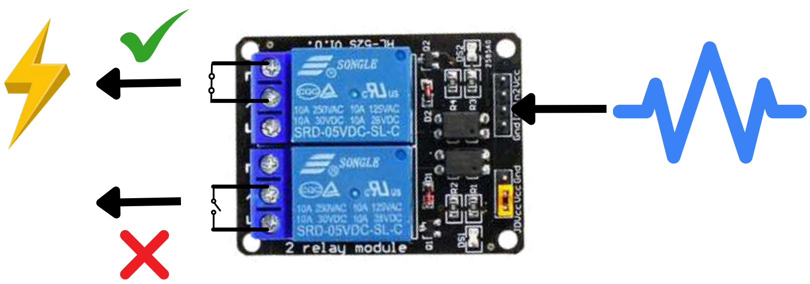
Main Parts :

- Relay Coils (×2): Create magnetism to open/close the switches.
- Transistor Driver: Boosts the weak control signal.
- Flyback Diode: Protects the circuit from voltage spikes.
- Screw Terminals: Connect the AC/DC load.

Working:

When the control pin gets a signal, the relay coil energizes and pulls the contacts, turning the connected device ON or OFF. Removing the signal makes the relay go back to its normal state.

Sensor Working



When signal applied to IN1, first relay coil energizes and current passes through the connected load. The other relay will not get energized

When signal applied to IN2, second relay coil energizes and current passes through the connected load. The other relay will not get energized.

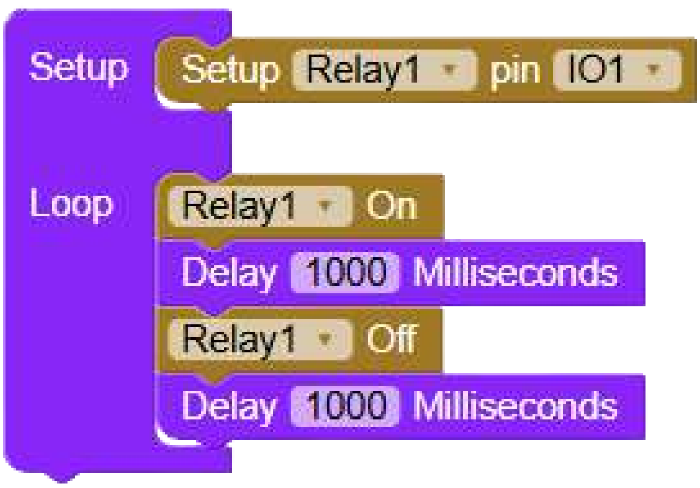
Models Compatible

Optocoupler based relay

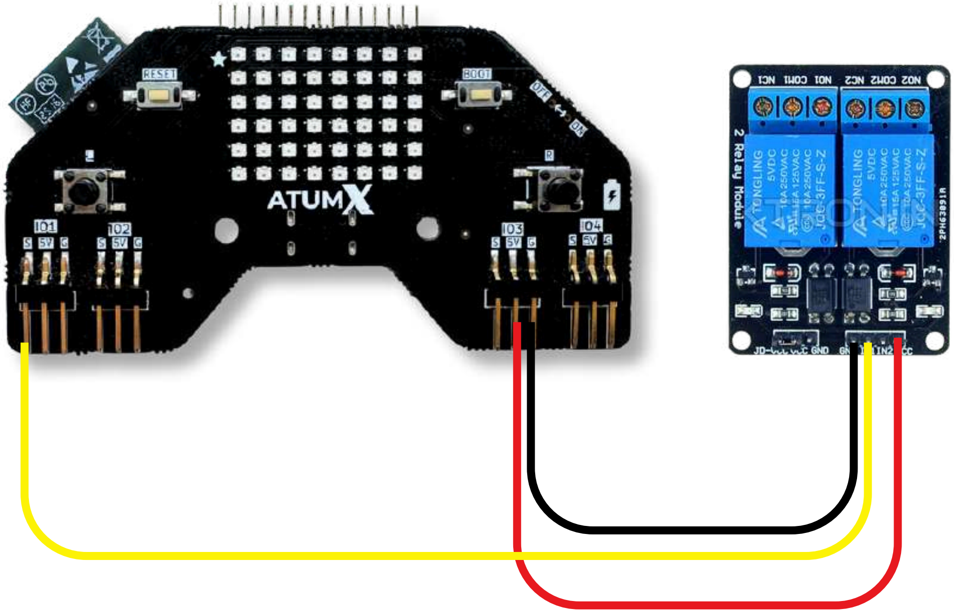
Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO1 and IO2 or any **available GPIO Pin** defined in the snippet block .	IN1

Block Code



Video link on how to connect



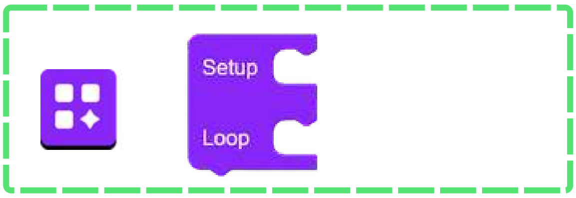
Power line

Relay

Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Setup Touch Sensor pin** and **Relay** block to assign the Touch sensor and Relay to **any GPIO pin** on the board.

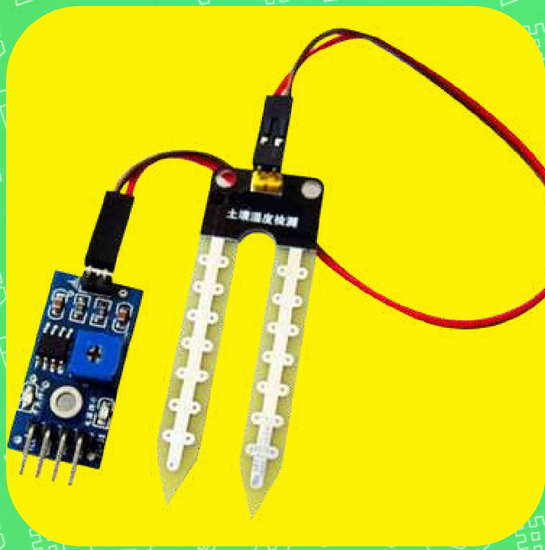
Block 3:

Use the **Relay block** to turn on or off the relay.

Block 4:

Use the **Delay** block to pause the execution for a while.

Soil moisture sensor



Know about soil moisture sensor

The Soil Moisture Sensor is used to measure how wet or dry the soil is. It helps microcontrollers like Arduino know when plants need water. The sensor provides an analog or digital output based on the moisture level in the soil.

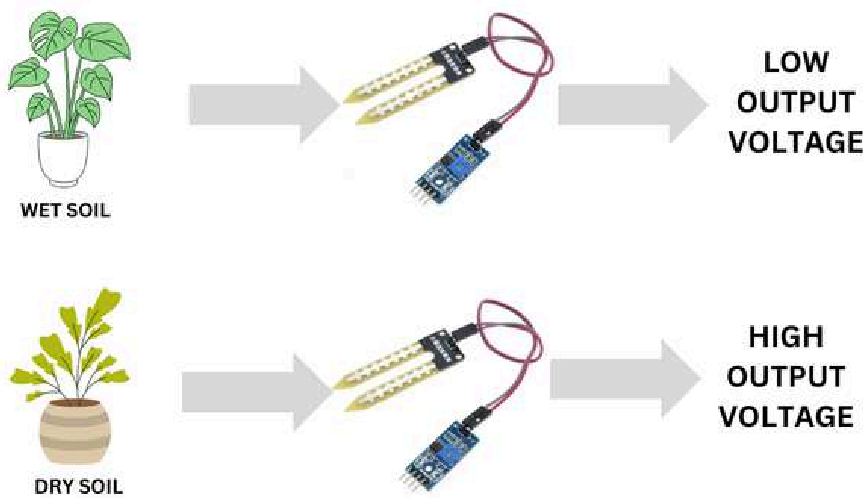
Main Parts:

- Probes/ Electrodes: Detect the water content in the soil.
- Comparator / Op-Amp (on module): Converts the probe reading into analog or digital signals.
- Indicator LEDs: Show whether soil is wet or dry.

Working Principle:

- Water in the soil increases conductivity between the probes.
- Dry soil has high resistance; wet soil has low resistance.
- The sensor reads this difference and outputs a signal that tells how much moisture is present.

Sensor Working



A soil moisture sensor works by measuring the electrical resistance between its probes, which changes based on how much water is present in the soil.

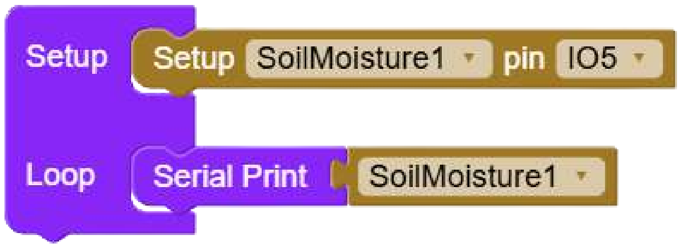
Models Compatible

FC-28 with LM293,
Capacitance based soil sensor

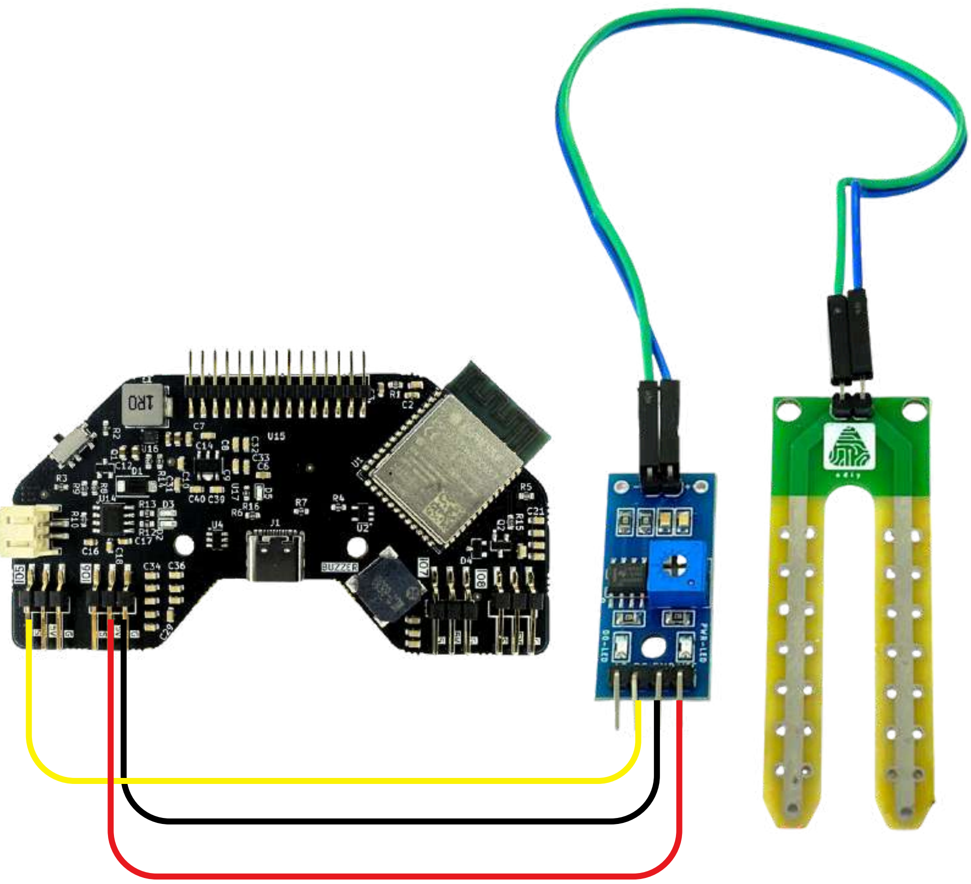
Pin Connection

Board	Sensor
In Board, connect to **any 5V Pins**	VCC
Connect to any **Ground pins**	GND
Connect to IO5 or any **available GPIO Pin** defined in the snippet block .	AO/DO

Block Code



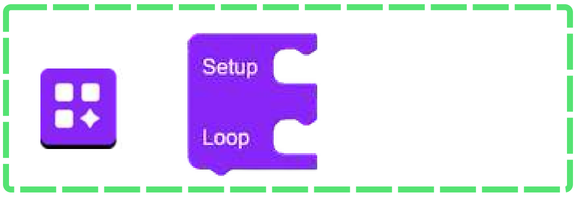
Video link on how to connect



Power line
DO
Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.

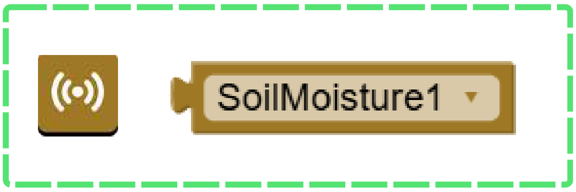


Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

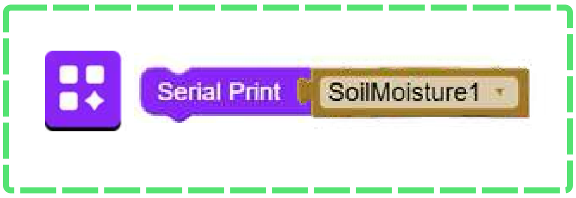
Block 2:

Configure the soil moisture sensor pins. Select any **available GPIO** for using this block



Block 3:

This block **reads the soil wetness or dryness.**



Block 4:

Use **Serial Print block** with soil moisture block to display soil moisture sensor values on the **Serial Monitor.**

DID YOU KNOW?

Soil moisture sensors help prevent overwatering by telling you exactly how much water the soil already has.

LCD Display

Know about LCD

The LCD (16x2) Display is used to show text, numbers, or symbols in electronic projects. It can display 16 characters per line and has 2 lines, making it useful for messages, sensor values, and menus.

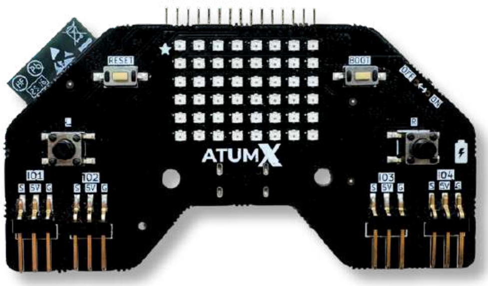
Main Parts:

a. LCD Panel: The screen where characters appear.
b. Controller Chip (HD44780): Converts microcontroller commands into characters on the display.
c. Data Pins (D0–D7): Carry the information to be shown.
d. Control Pins (RS, RW, EN): Manage how data is sent.
e. Backlight & Contrast Control: Help make the text visible.

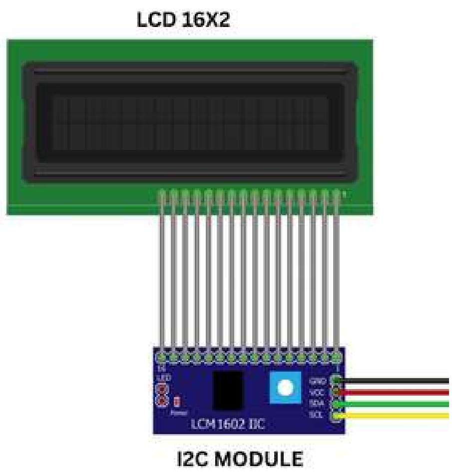
Working Principle:

The microcontroller sends data and commands to the LCD through the data and control pins.
The controller chip interprets these signals and updates the characters on the screen.
Backlight and contrast settings help you clearly view the displayed text.

Sensor Working



The board sends commands to LCD.



LCD displays the contents.

Models Compatible

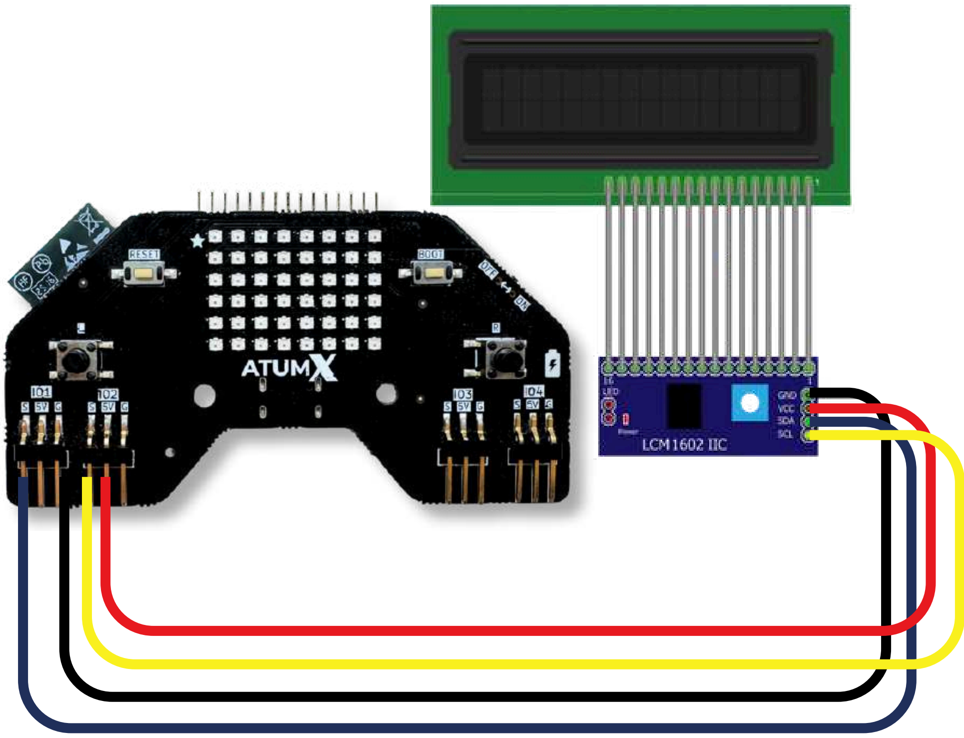
16x2 LCD1602 Display with I2C

Block Code

Pin Connection

Board	Sensor
5V	VCC
GND	GND
Connect to **S(IO1)** or any other **available IO pin** defined in the snippet block.	SCL
Connect to **S(IO2)** or any other **available IO pin** defined in the snippet block.	SDA

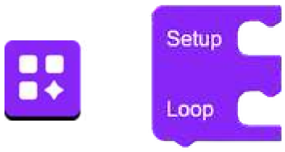
Video link on how to connect



Power line
SDA
SCL
Ground line

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Configure the LCD pins. Select any **available GPIO** for using this block

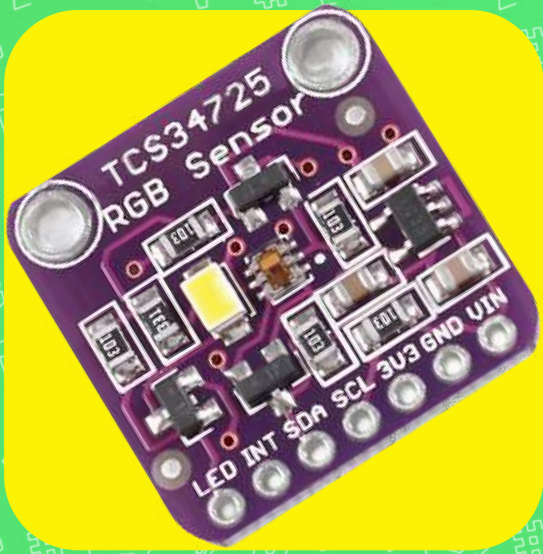
Block 3:

This block displays **Hello in LCD.**

DID YOU KNOW?

LCD screens don't produce their own light — they need a backlight to make the display visible.

Colour sensor



Know about colour sensor

The Color Sensor is used to detect the color of an object by measuring the amount of Red, Green, and Blue (RGB) light reflected from it. It sends this data to a microcontroller for color identification.

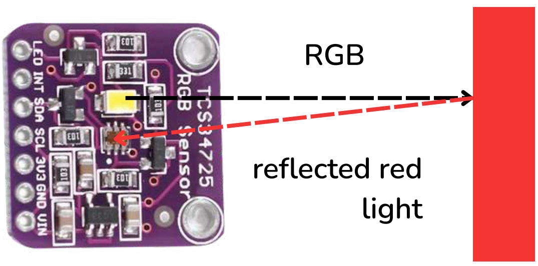
Main Parts:

- RGB Filters: Allow only red, green, or blue light to pass.
- Photodiodes: Convert the filtered light into electrical signals.
- Onboard Converter (Frequency/ Digital Output): Converts light intensity into digital or frequency signals.
- LEDs (on module): Provide constant illumination for accurate color reading.

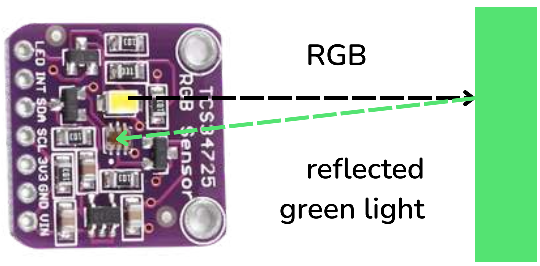
Working Principle:

- The sensor shines light on an object and measures how much red, green, and blue light is reflected.
- Each color filter activates one set of photodiodes.
- The sensor processes these readings and sends the RGB values to the microcontroller to determine the object's color.

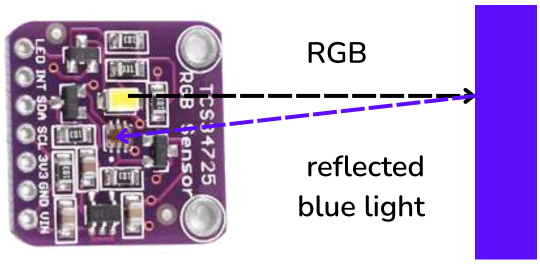
Sensor Working



When RGB light flashes on the object, red gets reflected and other lights get absorbed. The amount of reflected red light determines the voltage produced by the sensor.



When RGB light flashes on the object, green gets reflected and other lights get absorbed. The amount of reflected green light determines the voltage produced by the sensor.



When RGB light flashes on the object, blue gets reflected and other lights get absorbed. The amount of reflected blue light determines the voltage produced by the sensor.

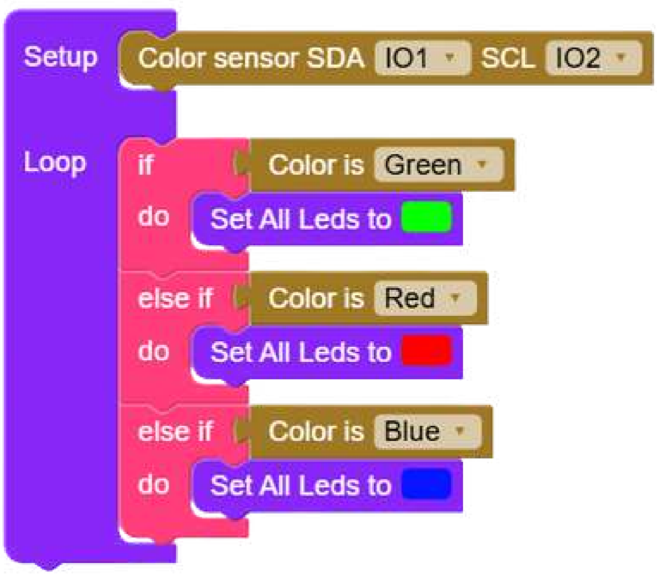
Models Compatible

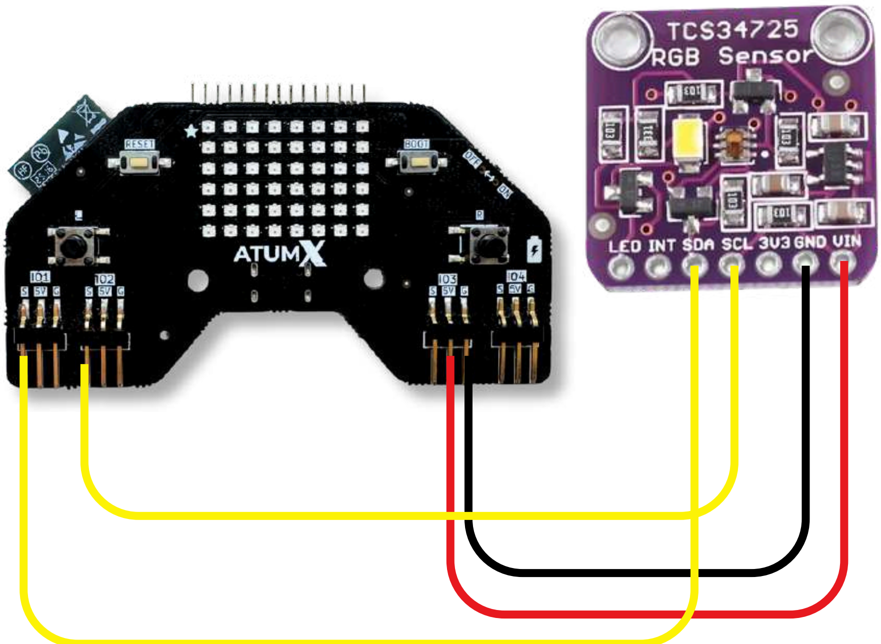
TCS34725

Pin Connection

Board	Sensor
5V	VCC
GND	GND
Connect to **S(IO1)** or any other **available IO pin** defined in the snippet block.	SDA
Connect to **S(IO2)** or any other **available IO pin** defined in the snippet block.	SCL

Block Code





Video link on how to connect

Power line

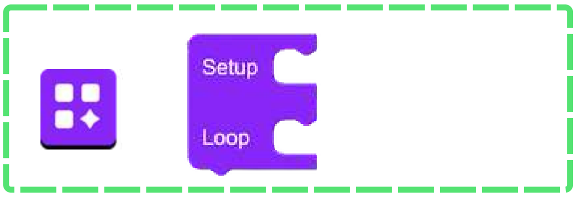
Trigger

Echo

Ground line

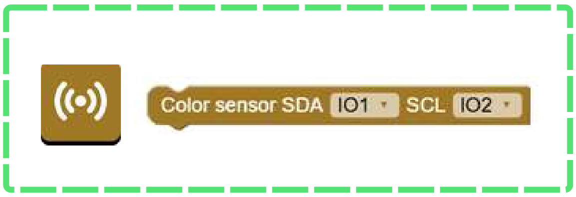
Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



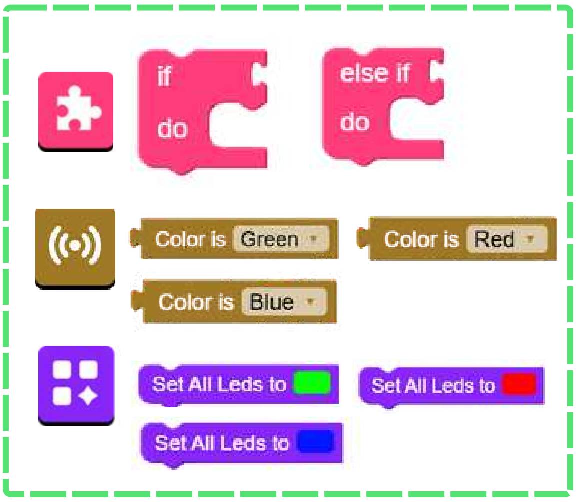
Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**



Block 2:

Use the **Setup Color Sensor pin** to assign the Color sensor to **any GPIO pin** on the board.



Block 3:

Use **If else** block attached with conditions to esure that if the color sensor detects **green** then the subo will glow in **green** color. Else if the color sensor detects **red** then the subo will glow in **red** color or if the color sensor detects **blue** then the subo will glow in **blue** color.

OLED Display

The OLED Display is a small screen that shows text, graphics, or symbols using tiny light-emitting pixels. It does not need a backlight, so it gives bright, clear output even in low light.

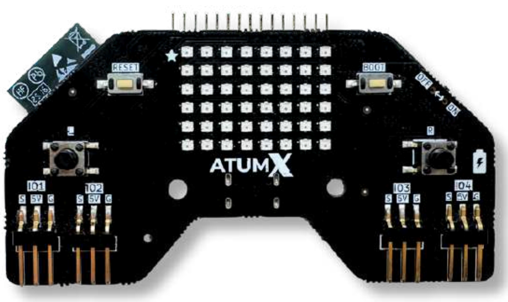
Main Parts:

- OLED Pixels: Tiny LEDs that light up to form images or text.
- Driver/Controller Chip (like SSD1306): Converts microcontroller commands into pixel data.
- Communication Pins (I^2C): Send data from the microcontroller to the display.
- Power Circuit: Powers the display and pixels.

Working Principle:

- The microcontroller sends commands through I^2C.
- The controller chip turns ON or OFF specific OLED pixels based on this data.
- These glowing pixels create the characters or graphics seen on the screen.

OLED Working



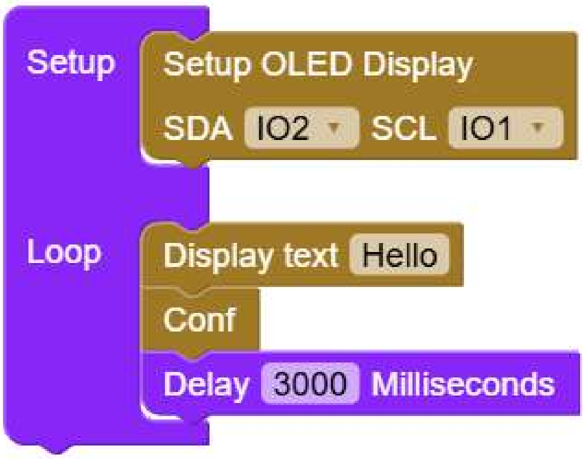
The board sends commands to OLED.

OLED displays the contents.

Models Compatible

1.3 inch (I2C) OLED display

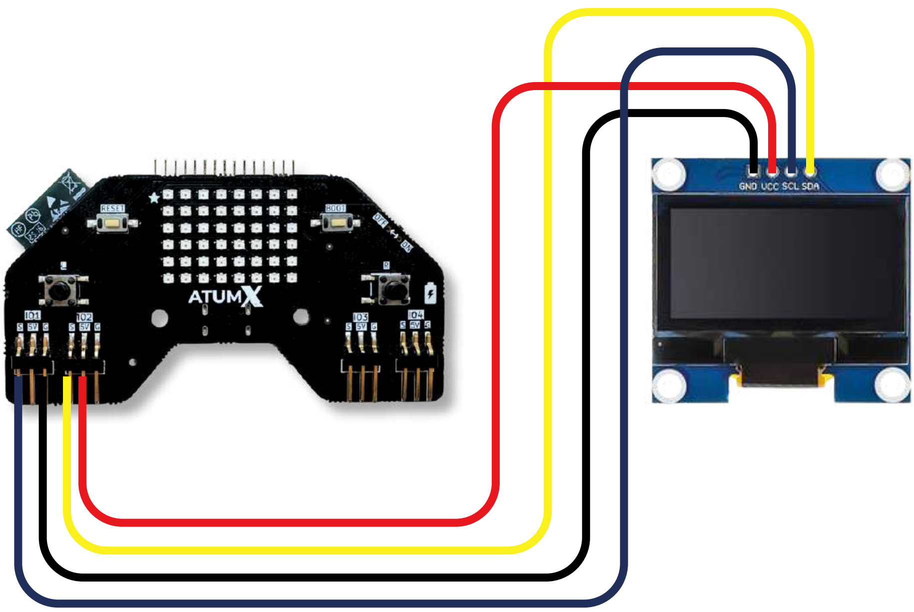
Block Code



Pin Connection

Board	Sensor
5V	VCC
GND	GND
Connect to **S(IO1)** or any other **available IO pin** defined in the snippet block.	SCL
Connect to **S(IO2)** or any other **available IO pin** defined in the snippet block.	SDA

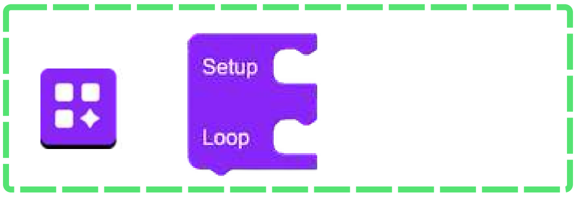
Video link on how to connect



Power line
SDA
SCL
Ground line

Check the IO pins you connected in the board and in the code snippet are same.

Code explanation



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Configure the OLED pins. Select any **available GPIO** for using this block

Block 3:

This block displays **hello and confetti** on OLED.

Block 4:

This block delays the next block execution to 3 seconds.

DID YOU KNOW?

OLED displays light up each pixel on their own, so they show brighter colors and deeper blacks while using less power.

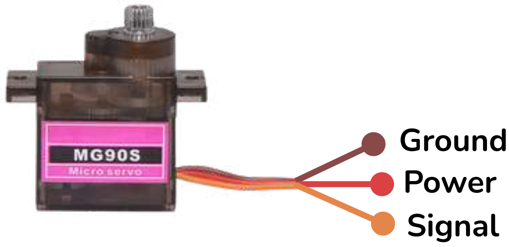
Servo Motor 180°

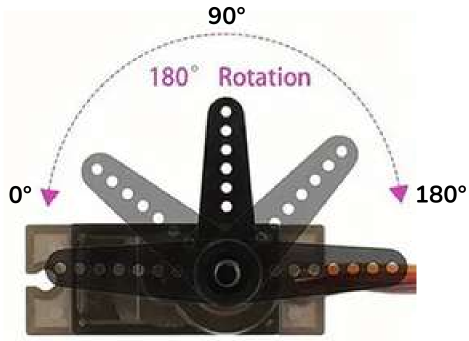
A 180° servo motor is a special type of motor that can rotate only within a fixed range — typically from 0° to 180°. You simply tell it the exact angle you want, and it turns to that position and stays there until the next command.

Inside, it contains a small DC motor connected to gears for slow, controlled movement. A built-in potentiometer continuously measures the shaft's angle and tells the control circuit whether it has reached the commanded position.

A 180° servo is controlled using PWM (Pulse Width Modulation) signals, where the width of the pulse represents the desired angle.

When you give a specific angle, the servo moves to that exact position, not beyond it. Sending the same angle again will not rotate it further — it will simply hold that angle.





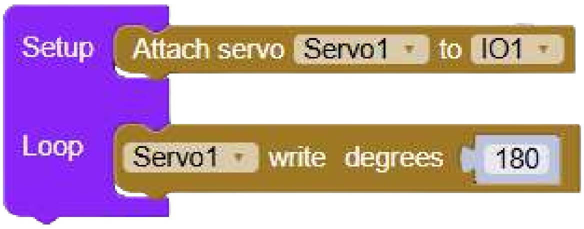
Servo motor can only move from 0 to 180°

Servo motor Working

Models Compatible

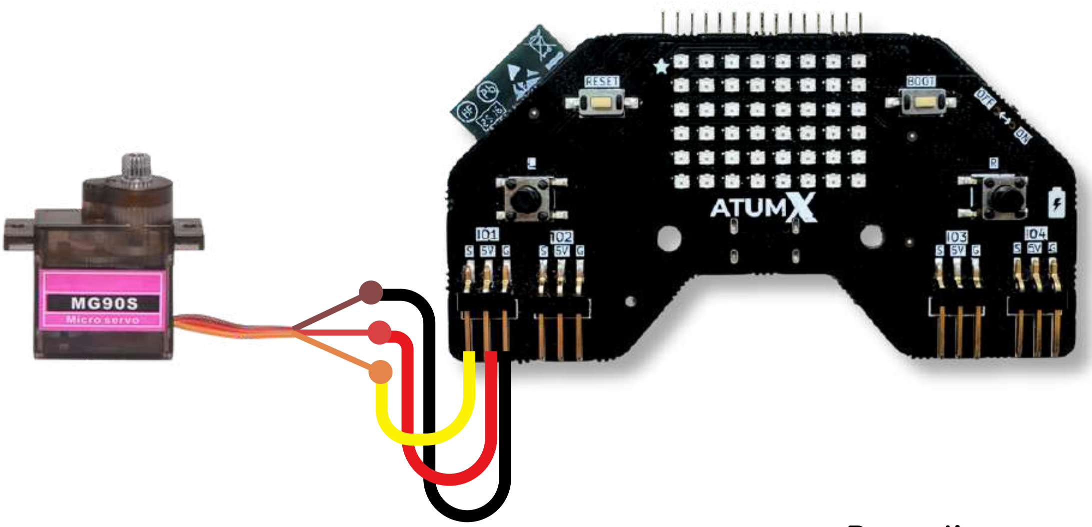
Tower Pro MG90 S Micro Gear
Servo Motor

Block Code



Pin Connection

Board	Sensor
Connect to IO1 or any S(IO) in board defined in the snippet block .	Yellow wire
5V	Red wire
Ground	Brown wire

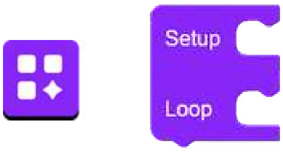


Power line
Ground line
signal

Video link on how
to connect

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Attach Servo** block to connect a servo motor to a specific pin on the board

Block 3:

Use the **Write Degrees** block to set the servo motor to a specific angle.

The servo motor moves to the particular angle that we have coded via blocks.

DID YOU KNOW?

Servos are tiny but powerful—they're used in airplanes to move control surfaces like rudders and flaps!

Servo Motor 360°

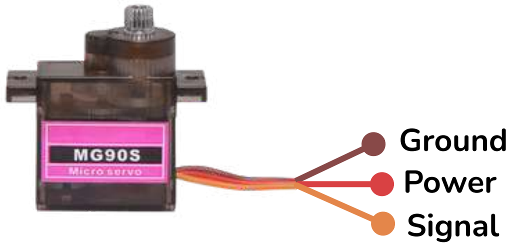
Know about Servo motor

A 360° servo motor, also called a continuous rotation servo, works differently from a 180° servo. Instead of moving to a specific angle, it can rotate full 360° and continue spinning like a regular DC motor — forward or reverse.

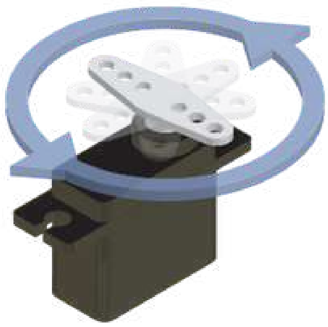
Inside, the potentiometer is replaced or modified so the servo no longer tracks position. Instead, the control circuit uses the PWM signal to control speed and direction, not angle.

A pulse near the center value stops the servo, a shorter pulse rotates it in one direction, and a longer pulse rotates it in the opposite direction.

Because it does not measure angle, you cannot tell a 360° servo to go to 90° or 150°. Instead, you instruct it to rotate clockwise or anticlockwise and at what speed.



Full 360 degrees rotation

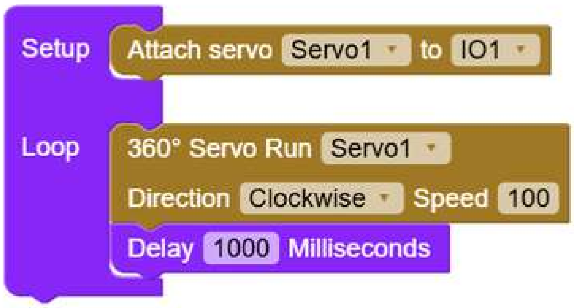


Servo motor Working

Models Compatible

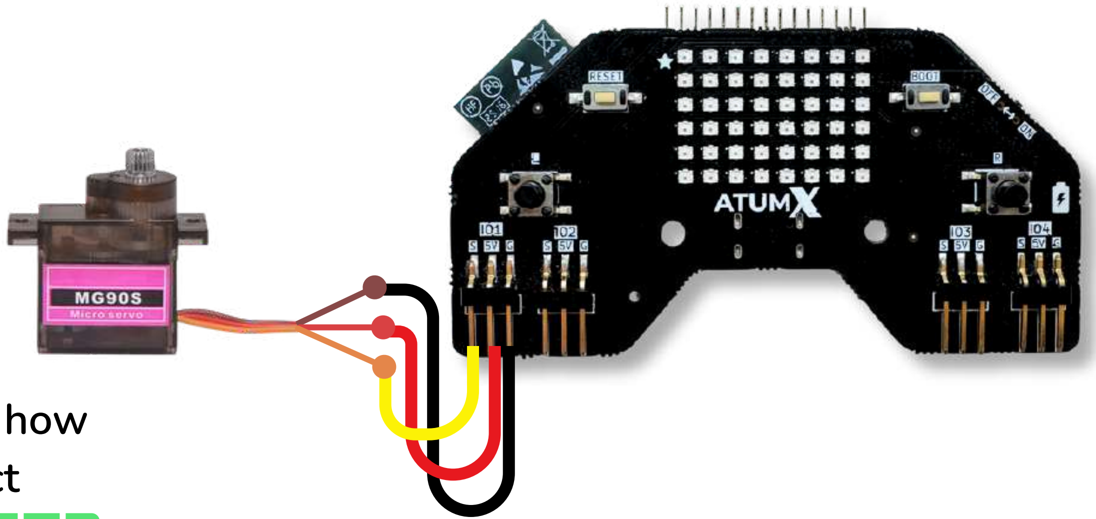
Tower Pro MG90 S Micro Gear
Servo Motor

Block Code



Pin Connection

Board	Sensor
Connect to IO1 or any S(IO) in board defined in the snippet block .	Yellow wire
5V	Red wire
Ground	Brown wire



Power line

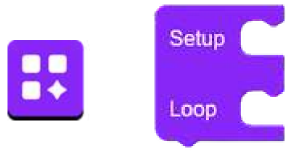
Ground line

signal

Video link on how to connect

Code explanation

Check the IO pins you connected in the board and in the code snippet are same.



Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

Use the **Attach Servo** block to connect a servo motor to a specific pin on the board

Block 3:

Use the **Write speed and direction** block to set the servo motor to a speed and direction.

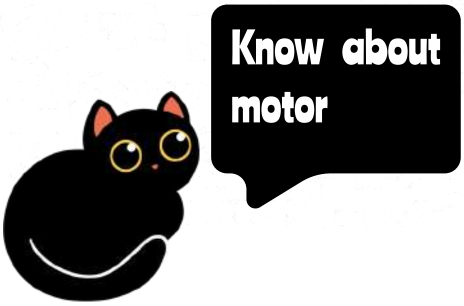
Block 4:

Use appropriate delay to stop the servo motor.

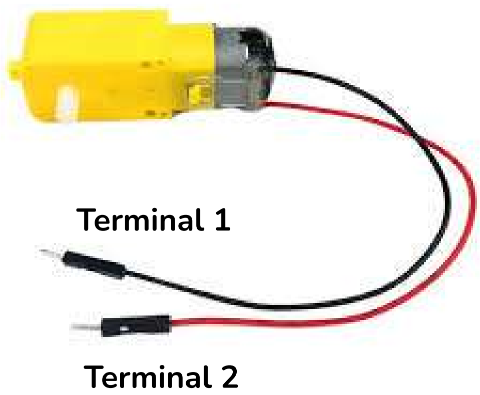
DID YOU KNOW?

Surgical robots use high-precision servos that can perform movements more accurate than a human hand.

Motor Magic



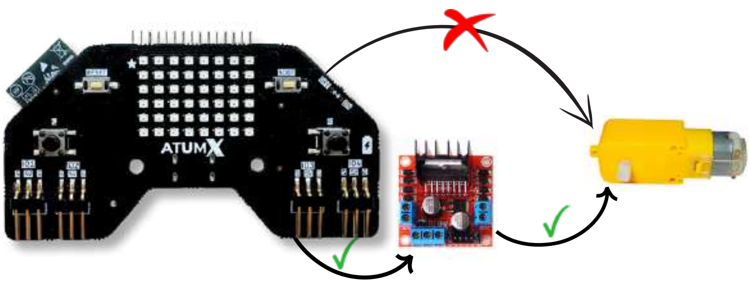
A **DC motor** (Direct Current motor) is a device that converts electrical energy into mechanical energy, producing **rotational motion.** When current flows through the motor, it creates a magnetic field that interacts with the motor's internal magnets, causing the shaft to spin. The speed and direction of rotation can be controlled by adjusting the voltage and polarity supplied to it.



Working Principle:

A DC motor works on the principle that a current-carrying conductor placed in a magnetic field experiences a force (Lorentz force). This force causes the motor's armature to rotate, converting electrical energy into mechanical motion.

Motor Working



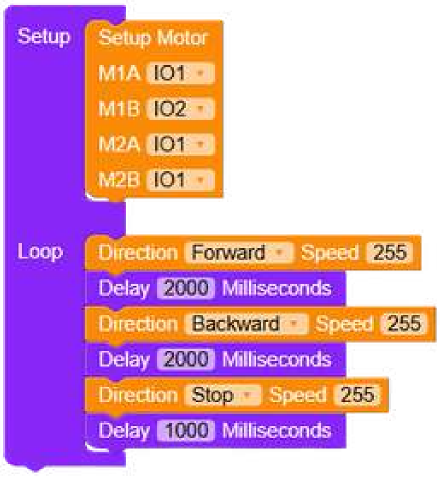
Connecting motor via motor controller

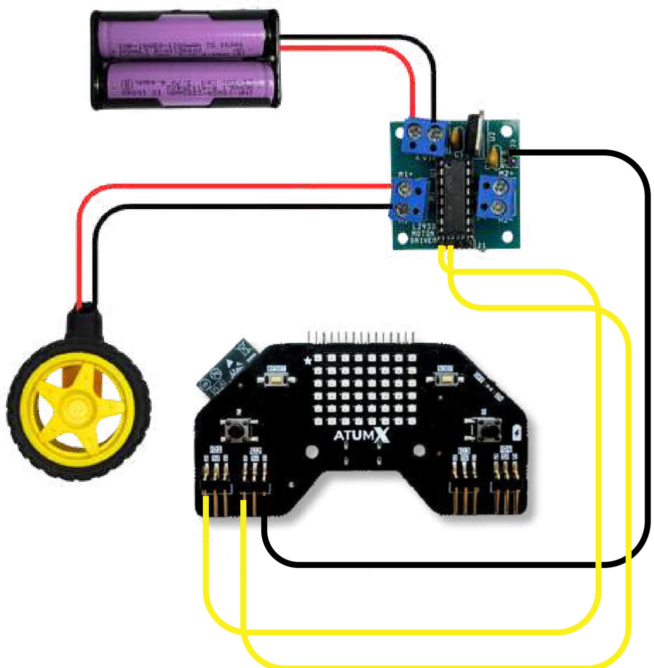
A motor controller manages the power and direction of motors, acting as an intermediary between the microcontroller and the motor. It ensures precise control over speed and direction, protects the motor and microcontroller, and handles the difference in voltage and current requirements.

Models Compatible

L293D / L298N Motor Driver Module

Block Code





Power line
IN1
IN2
Ground line

Pin Connection

Board	Motor Controller
Any GND	GND
IO1 or(any IO pin defined in the snippet block)	IN1
IO2 or (any IO pin defined in the snippet block)	IN2

Motor	Motor Controller
Wire 1	M1 +
Wire 2	M1 -

Double cell battery	Motor Controller
Red Wire	VIN
Black Wire	GND

If you want to connect our board with motor without using USB, Use a JST battery as shown below

Video link on how to connect

The motor will spin forward when one IN pin is HIGH and the other is LOW, and reverse when their states are swapped. If both IN pins are the same (both HIGH or both LOW), the motor will stop.

Check the IO pins you connected in the board and in the code snippet are same.

Code explanation

Block 1:

From **Basics**, drag the Setup Loop block. Code in **Setup runs once** at the start, while code in **Loop runs continuously.**

Block 2:

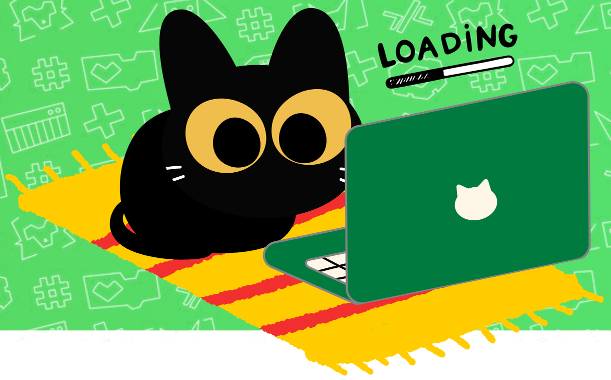
Set the corresponding **M1A** to **M2B** pins as per your motor connections with our board

Block 3:

Using the **Direction** block, set the desired direction that the motor needs to run along with the speed at which you want the motors to run.

Block 4:

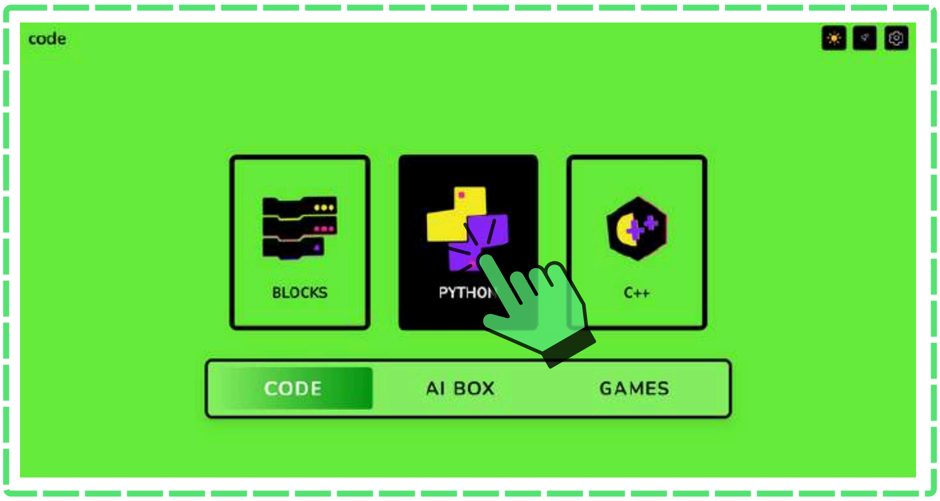
Use the **Delay** block to pause the execution for a while.



Blocky to python mode

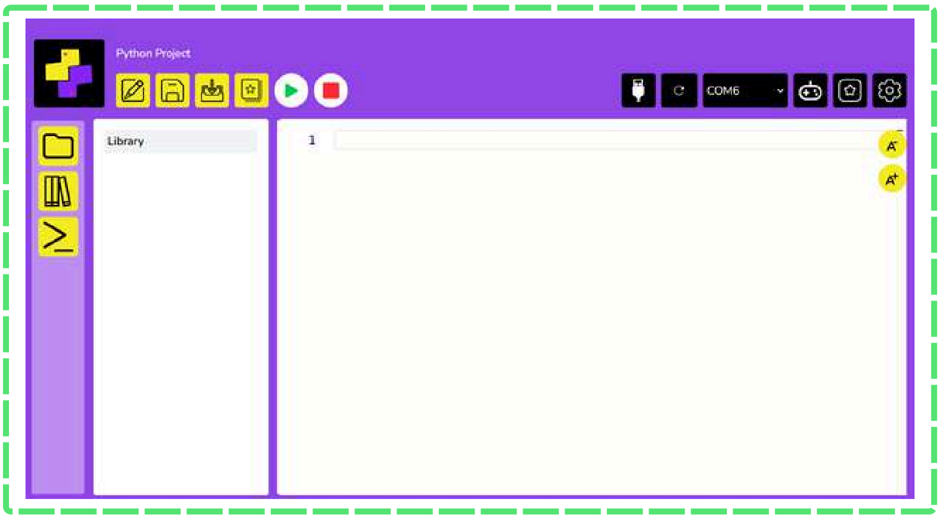
Step 1:

Click on **home button** in blocky mode.

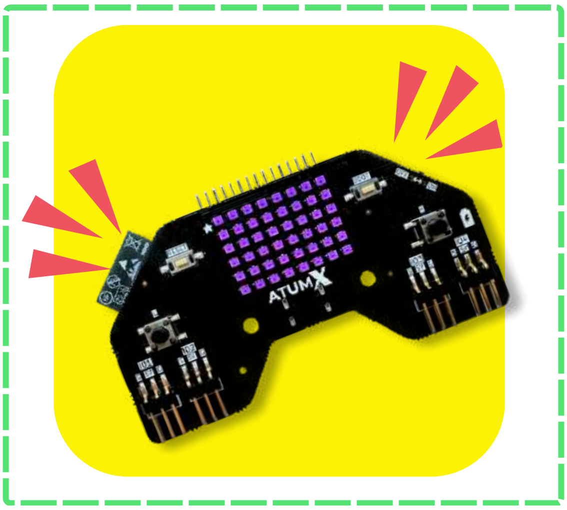


Step 2:

Click Python in code section and follow the steps specified in pg no. 7. The board will glow in purple color.



Now the app is in python mode.



Board LEDs glow in purple indicating a successful switch from blockly to python mode!

Python to blocky mode

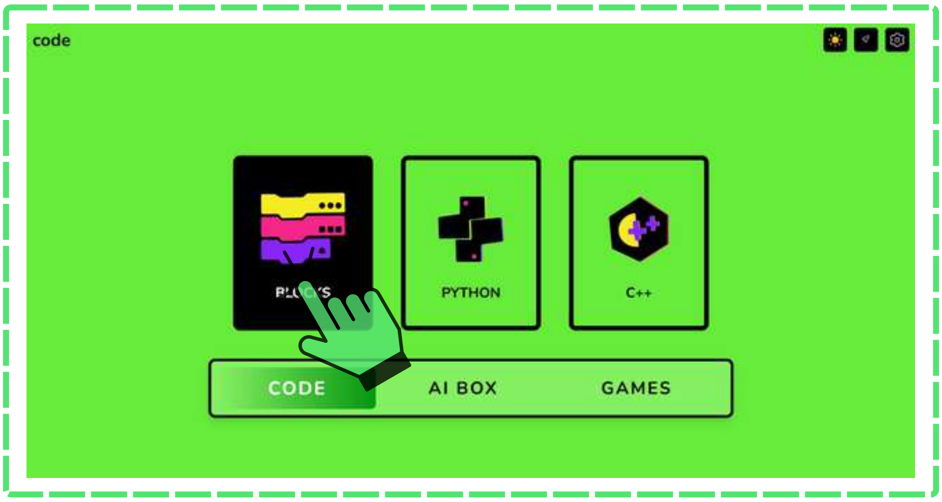
Step 1:

Click on **home button** in python mode.

Step 2:

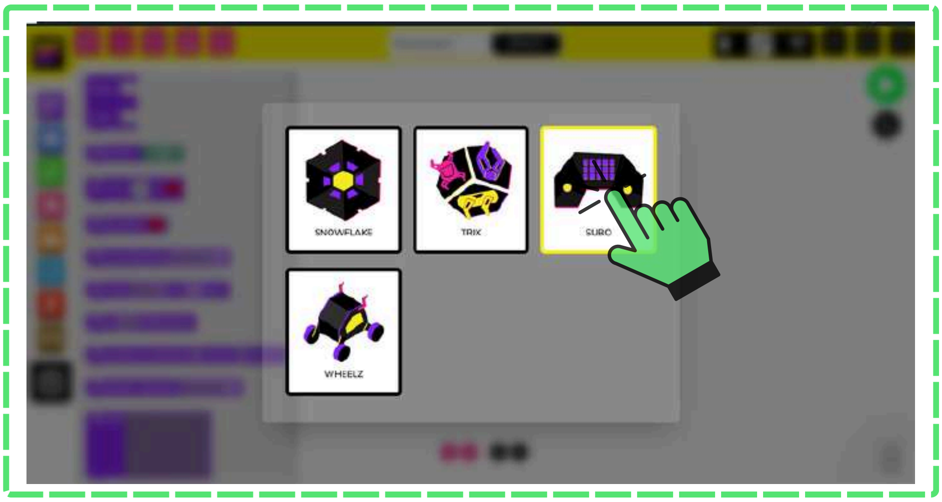
Click **Reset** button in the board after the blue light blink.

Once resetting the board, click **ok** on the popup displayed.



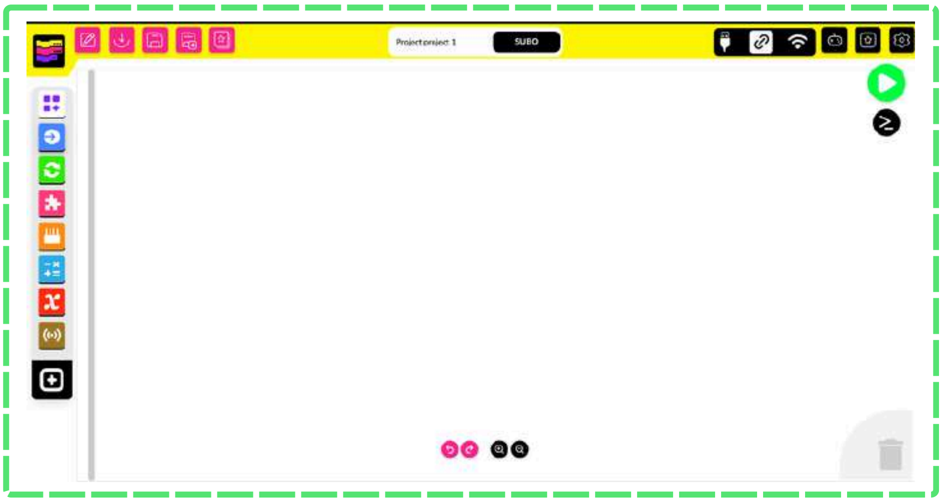
Step 3:

Click on the App you have downloaded and press Blocks in code section.

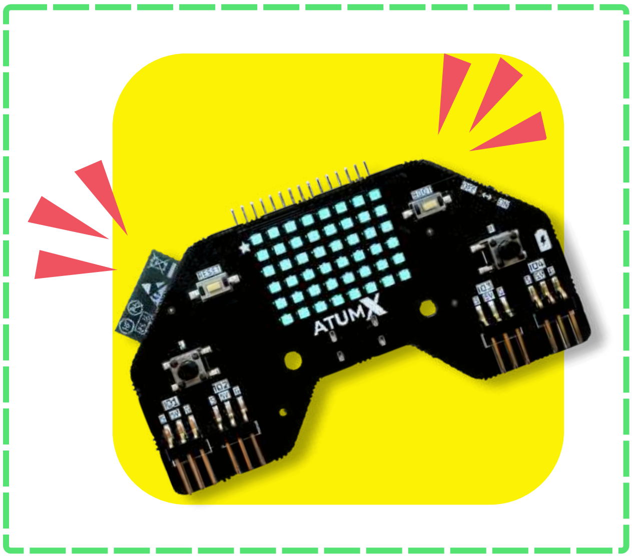


Step 4:

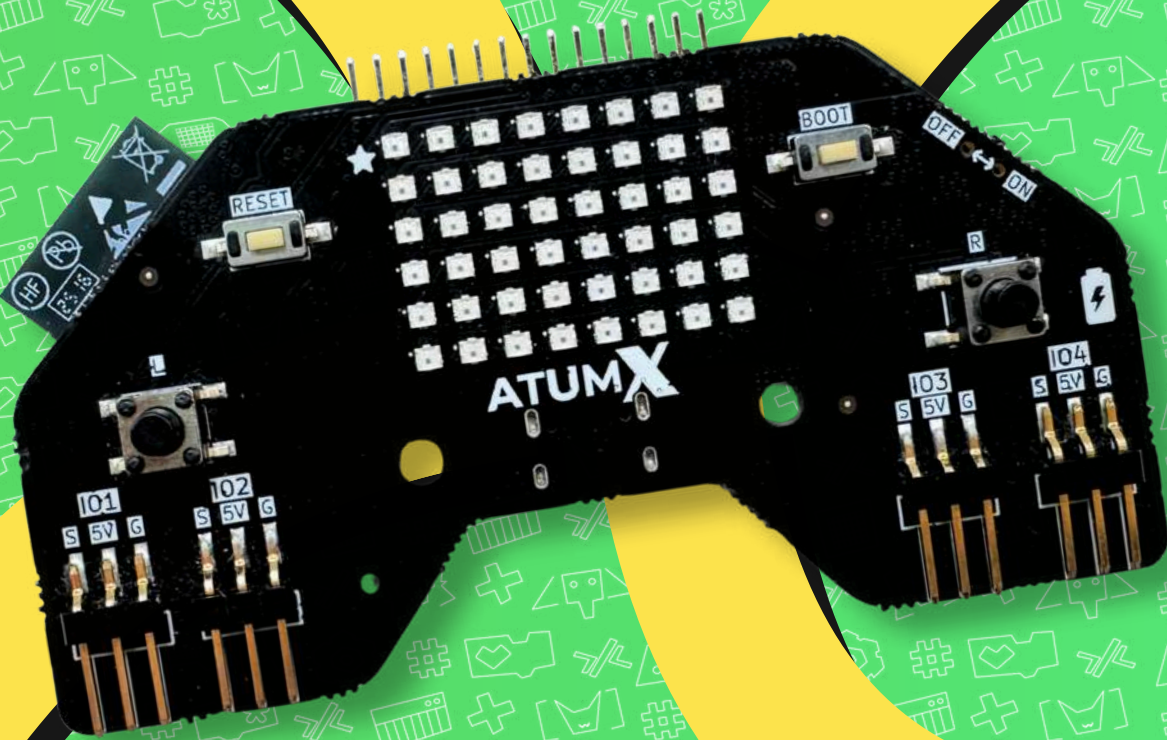
Click on Subo.



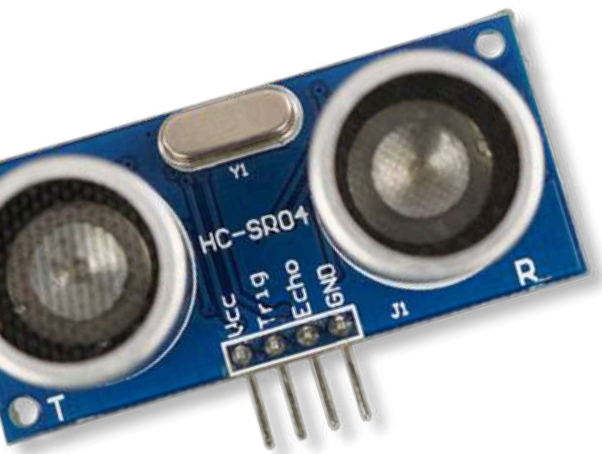
The app is now opened and ready for coding.



Board LEDs glow in blue indicating a successful switch from python to blockly mode!


RESET
BOOT
OFF
ON
R
L
ATUMX
IO1
S 5V G
IO2
S 5V G
IO3
S 5V G
IO4
S 5V G

See YOU SOON!


HC-SR04
Vcc Trig Echo GND
T
R